

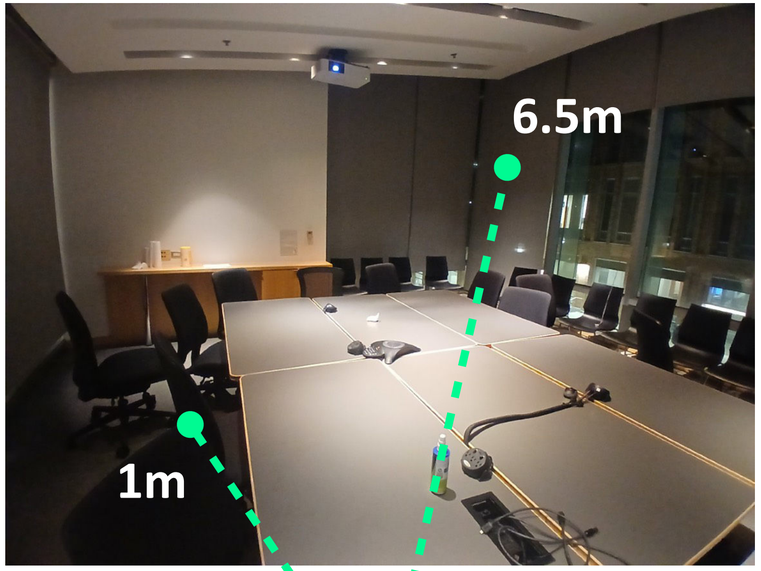
SHED Light on Segmentation for Depth Estimation

Presentor: Seung Hyun Lee
Advisor: Professor Stella X. Yu
Preliminary Exams

Committee:
Professor Anhong Guo
Professor Honglak Lee
Professor Nikola Banovic

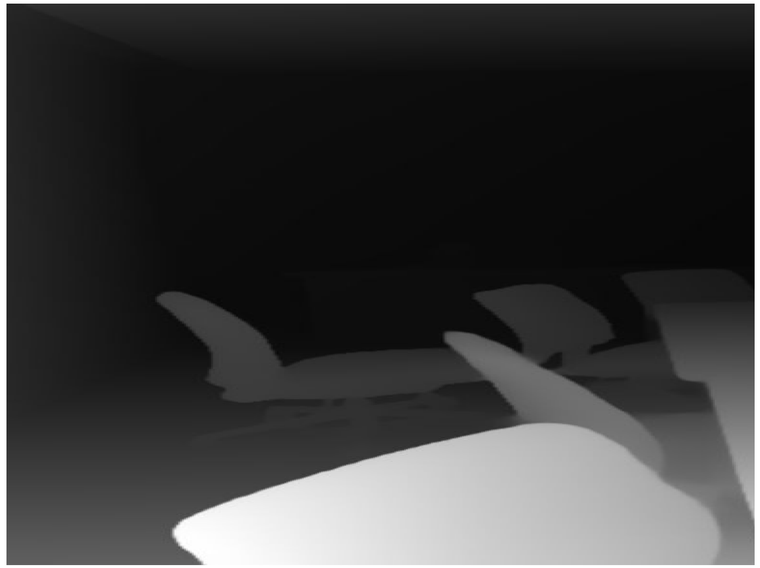
What is Depth?

Scene



RGB-D camera

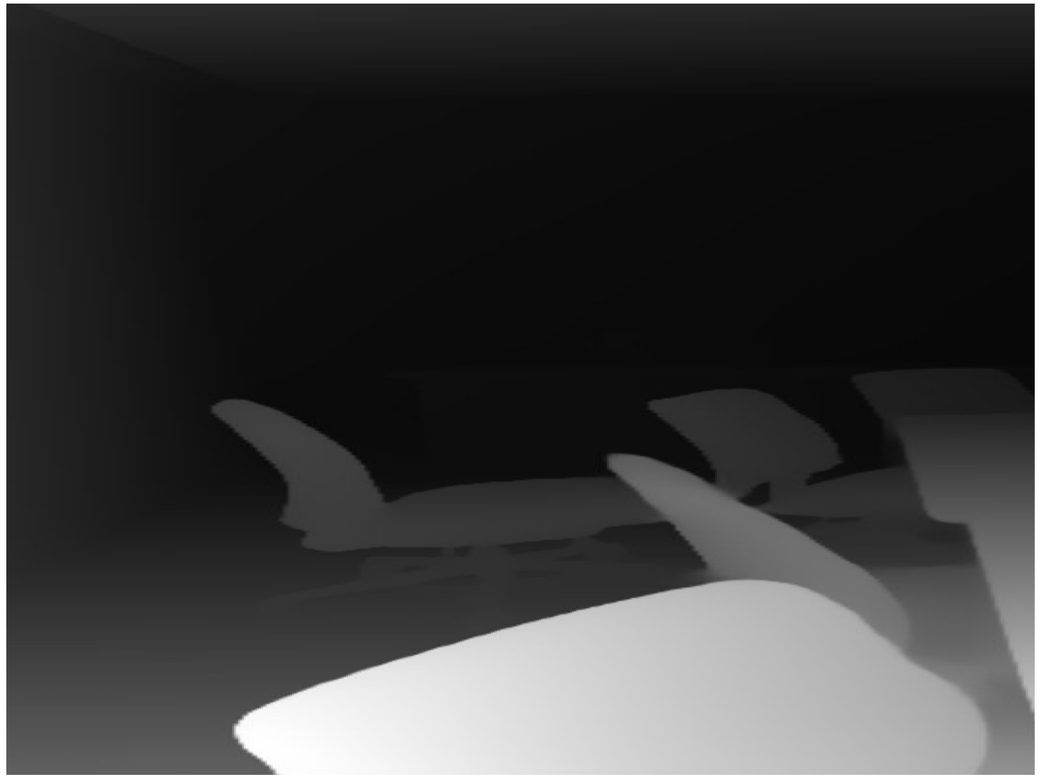
Depth (dense) map



Absolute Depth

Pixel value = Physical distance





Far (< 10m)

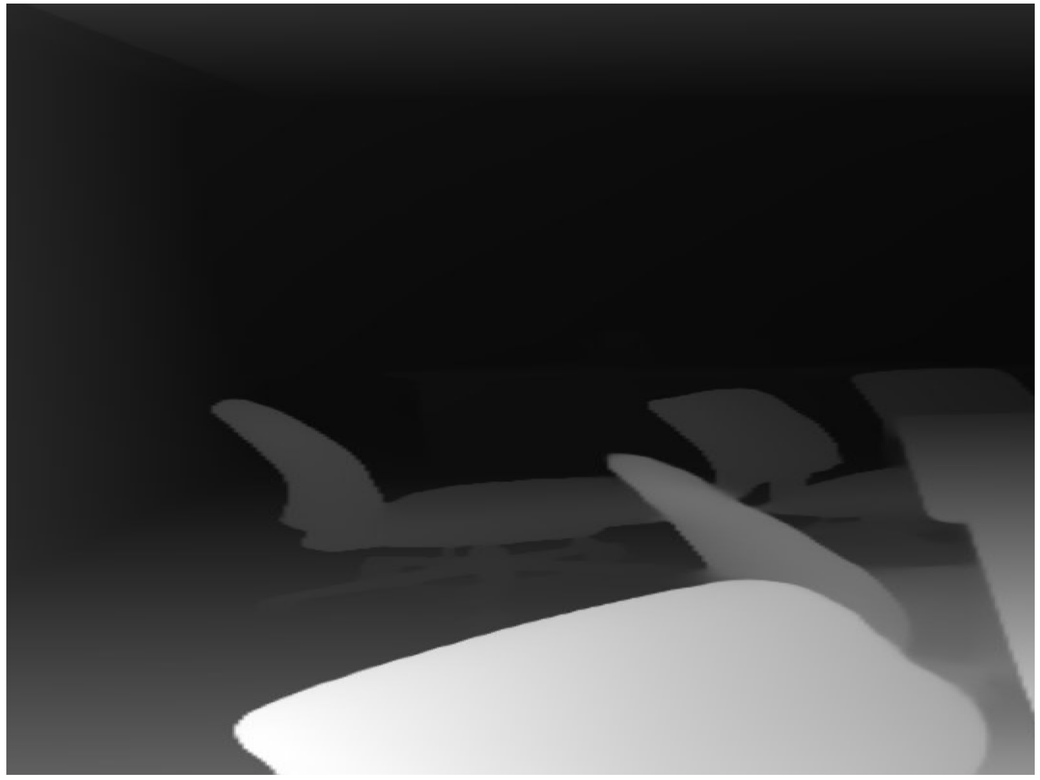


Near (> 0m)

Relative Depth

Which points are closer?





Far (> 0)

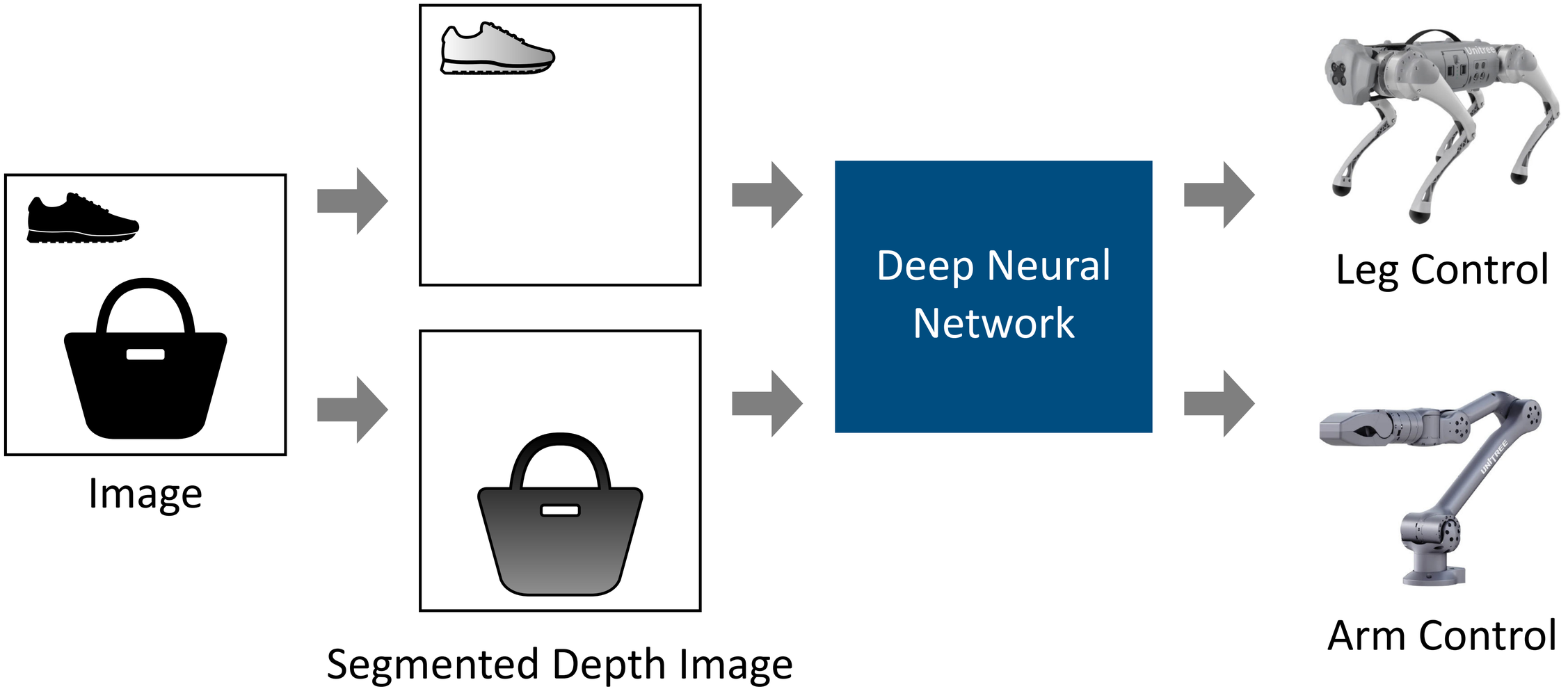
Near (< 1)

Application 1. Robot control for manipulation



Liu, Minghuan, et al. Visual whole-body control for legged loco-manipulation (CoRL 24)

Application 1. Robot control for manipulation



Liu, Minghuan, et al. Visual whole-body control for legged loco-manipulation (CoRL 24)

Application

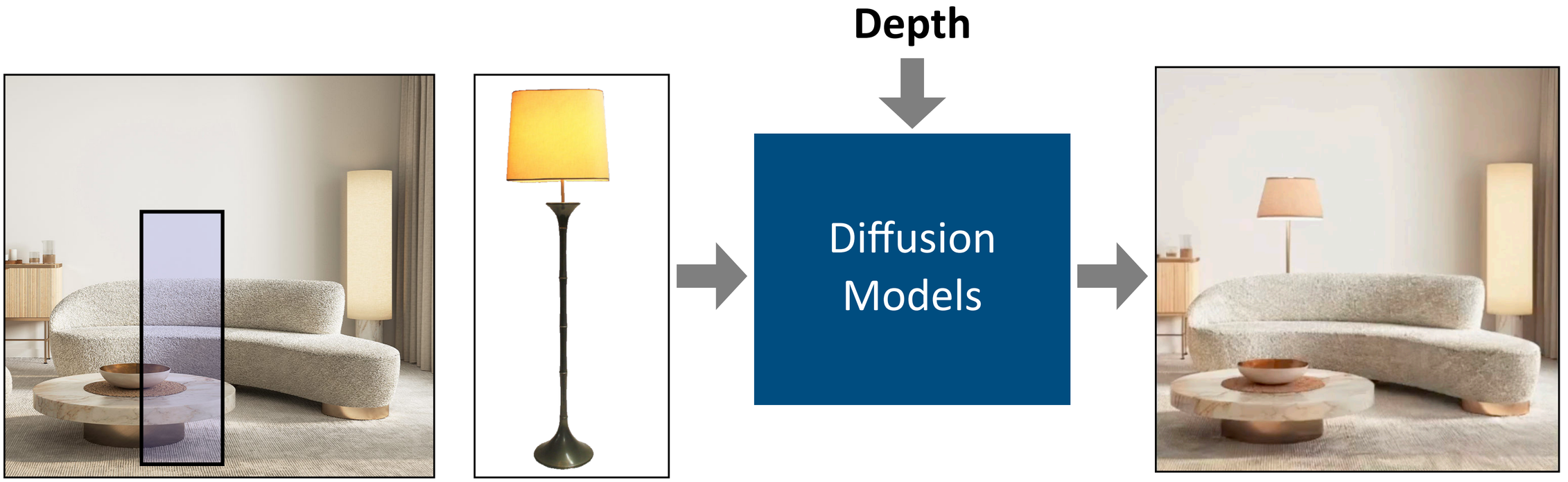
1. Robot control for manipulation
2. Autonomous driving



Zheng, Jingxiao, et al. "Multi-modal 3d human pose estimation with 2d weak supervision in autonomous driving. (CVPR 22)

Application

1. Robot control for manipulation
2. Autonomous driving
3. Depth-aware image editing



Rishubh, Parihar, et al. "Zero-Shot Depth-Aware Image Editing with Diffusion Models" (ICCV 25)

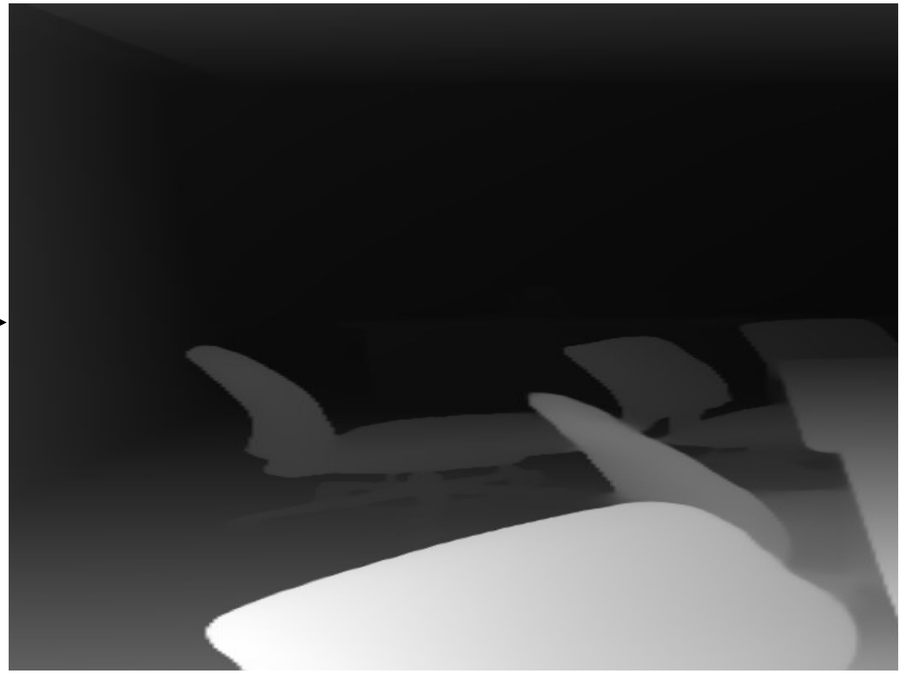
Monocular Depth Estimation (MDE)

Input image

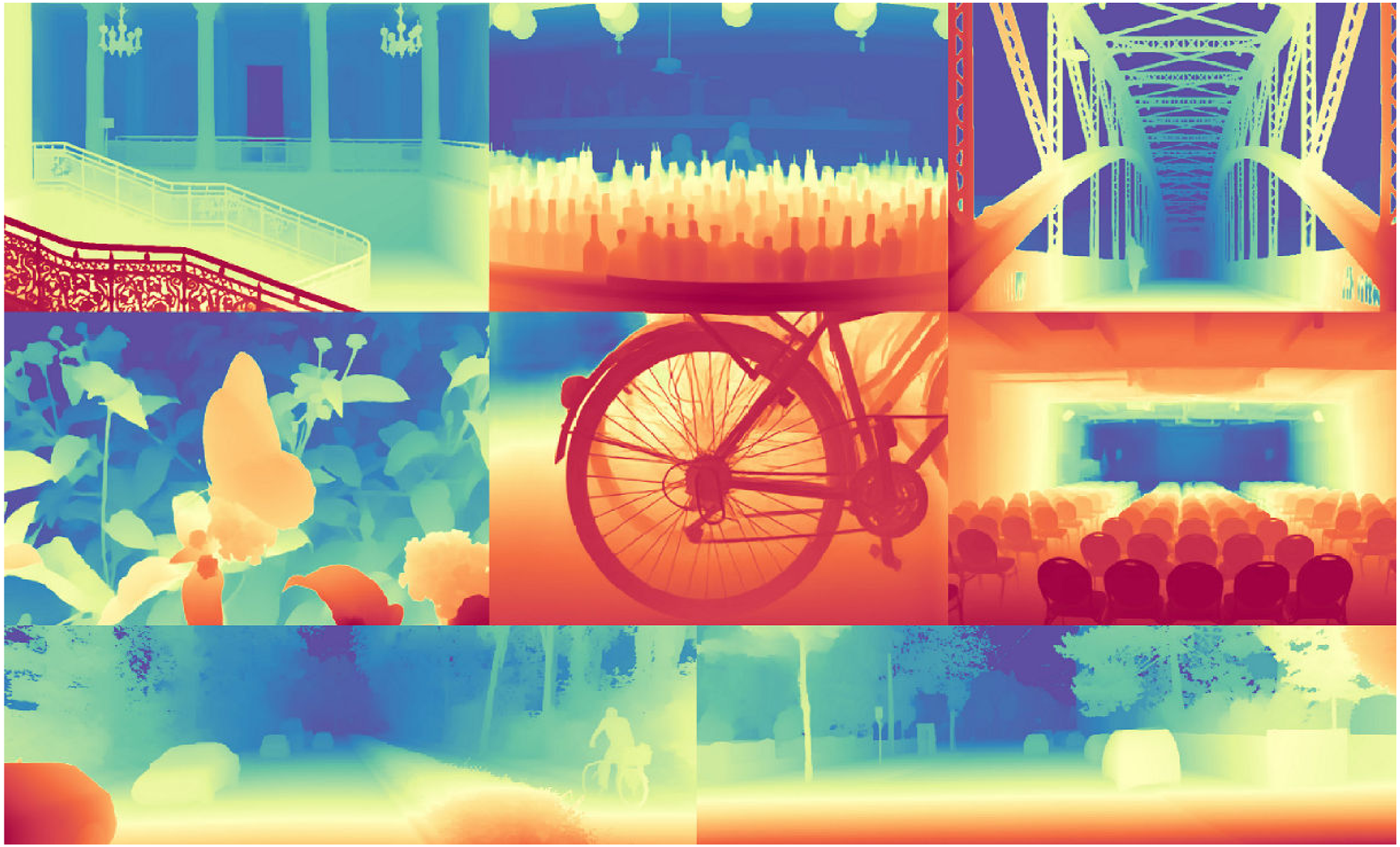


Neural Network

Depth (dense) map



Related Works 1. Depth Anything 2



Yang, Lihe, et al. "Depth anything v2." NeurIPS (2024)

Related Works

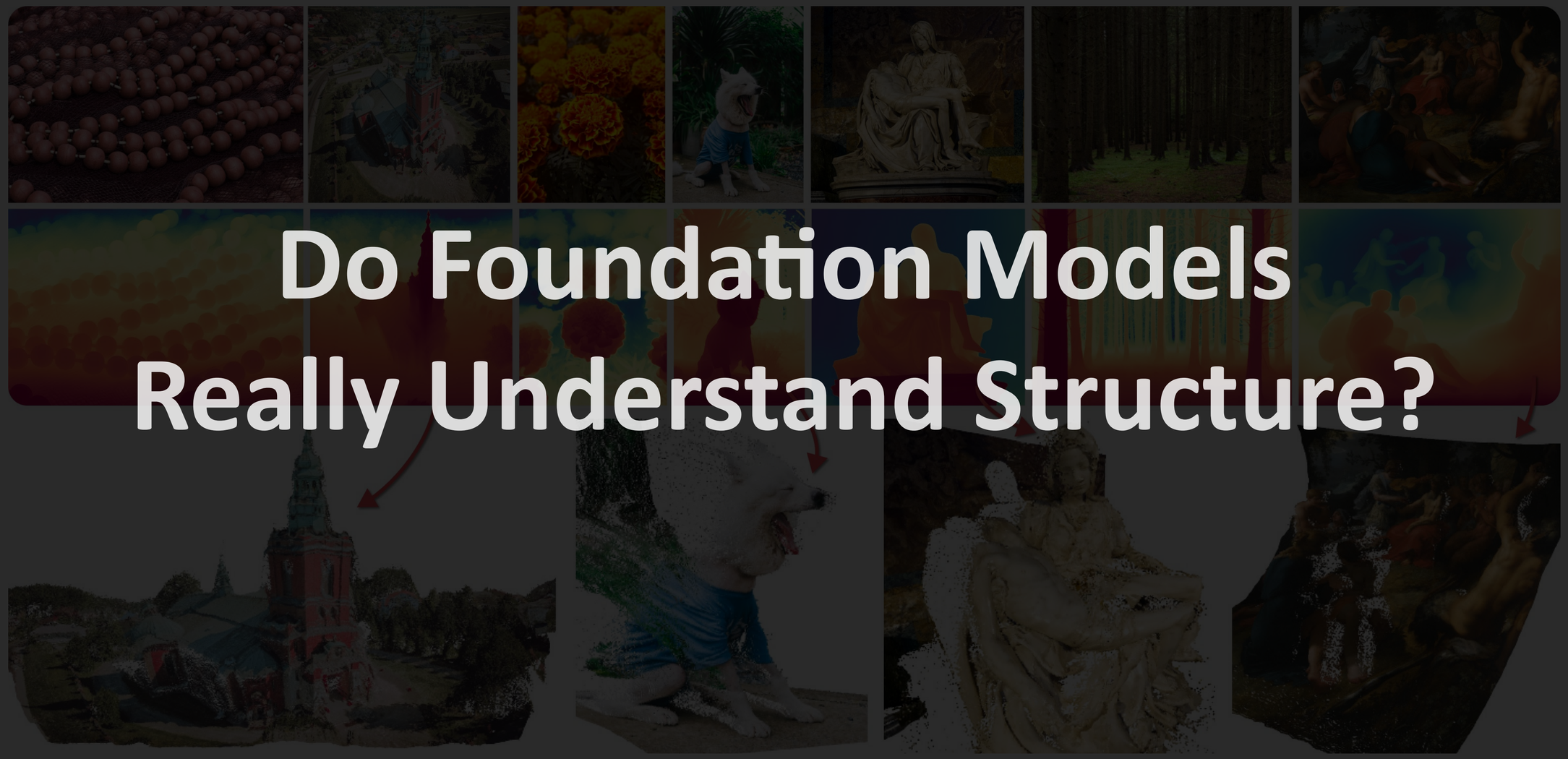
1. Depth Anything 2
2. Marigold



Rishubh, Parihar, et al. "Zero-Shot Depth-Aware Image Editing with Diffusion Models" (ICCV 25)

Related Works

1. Depth Anything 2
2. Marigold



Do Foundation Models Really Understand Structure?

Rishubh, Parihar, et al. "Zero-Shot Depth-Aware Image Editing with Diffusion Models" (ICCV 25)

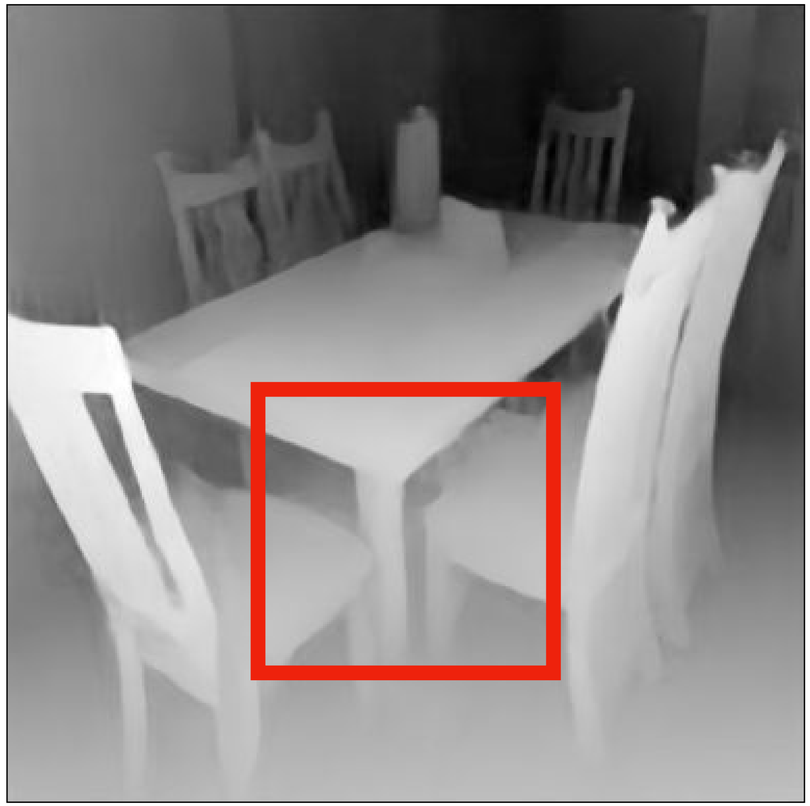
Motivation

1. Quality of depth map - blurry object boundaries

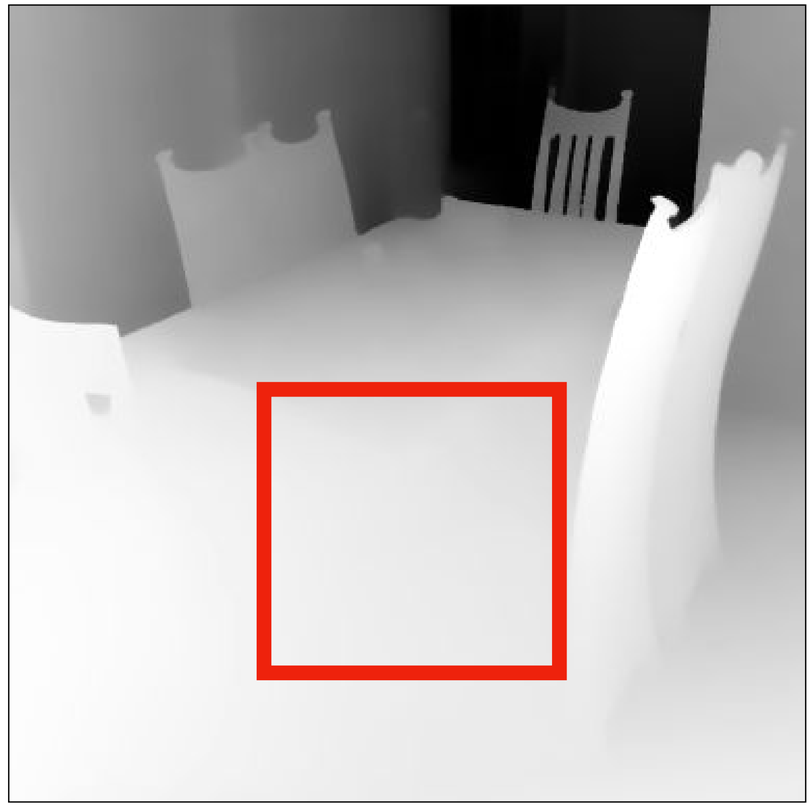
Input Image



Depth Anything 2



Marigold



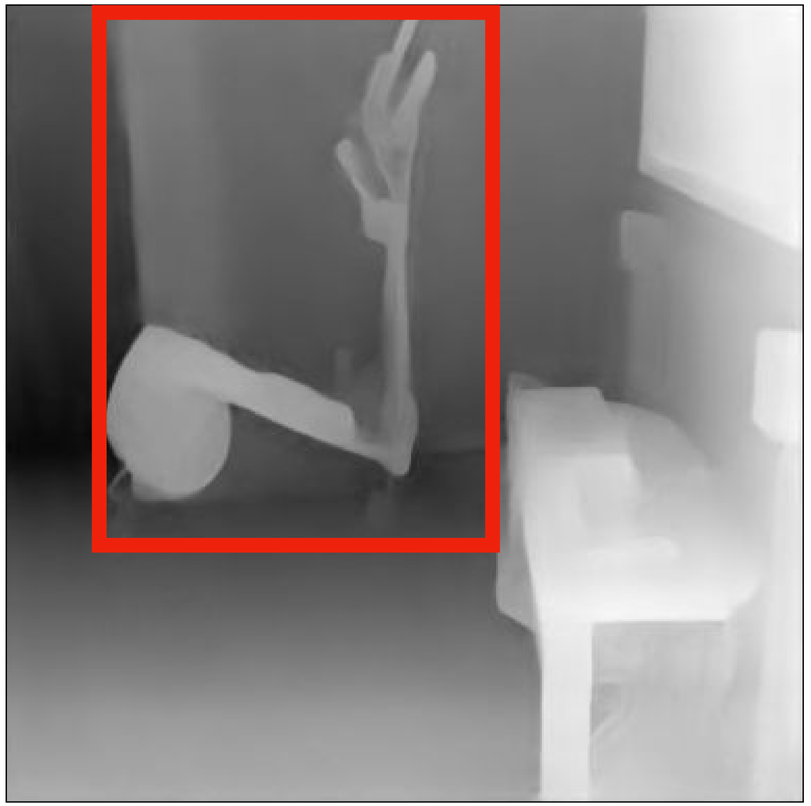
Motivation

1. Quality of depth map - blurry object boundaries
2. Quality of depth map - inconsistent depth values

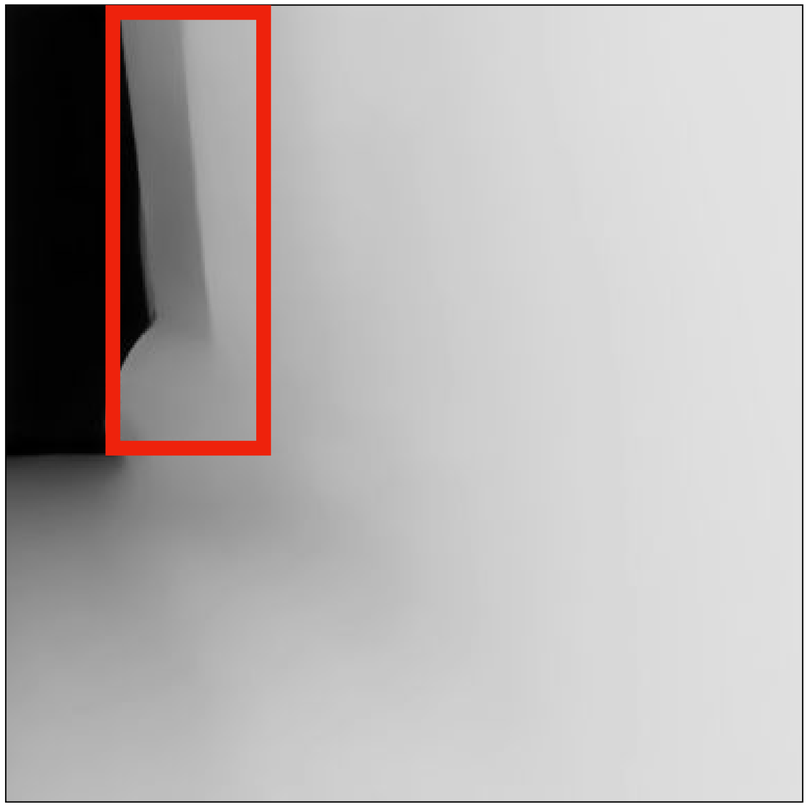
Input Image



Depth Anything 2



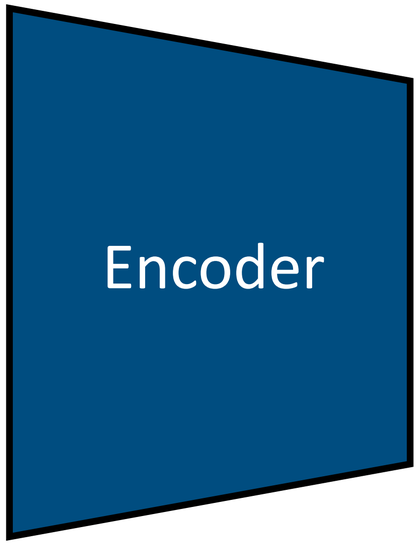
Marigold

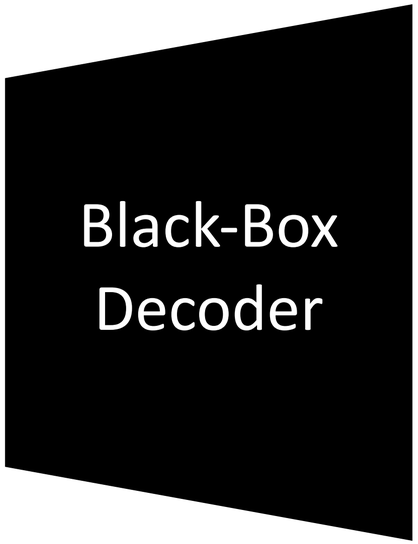


Motivation

1. Quality of depth map - blurry object boundaries
2. Quality of depth map - inconsistent depth values
3. Per-pixel prediction is not about modeling structure

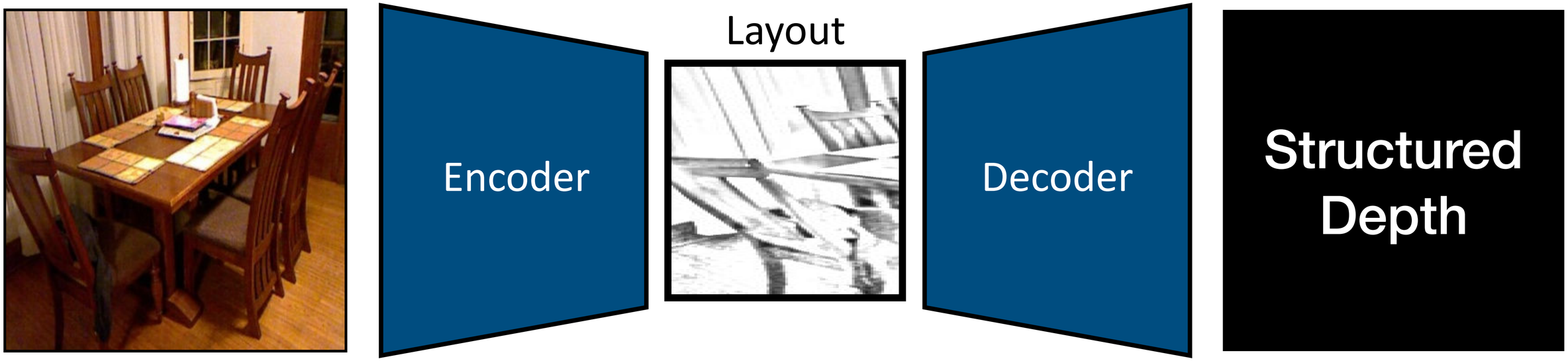








Modeling Structure for Structured Depth



Related Work

CAST: Part-to-whole discovery for image classification

Free internal segmentation without any supervision

Input Image



Fine Segments



Coarse Segments

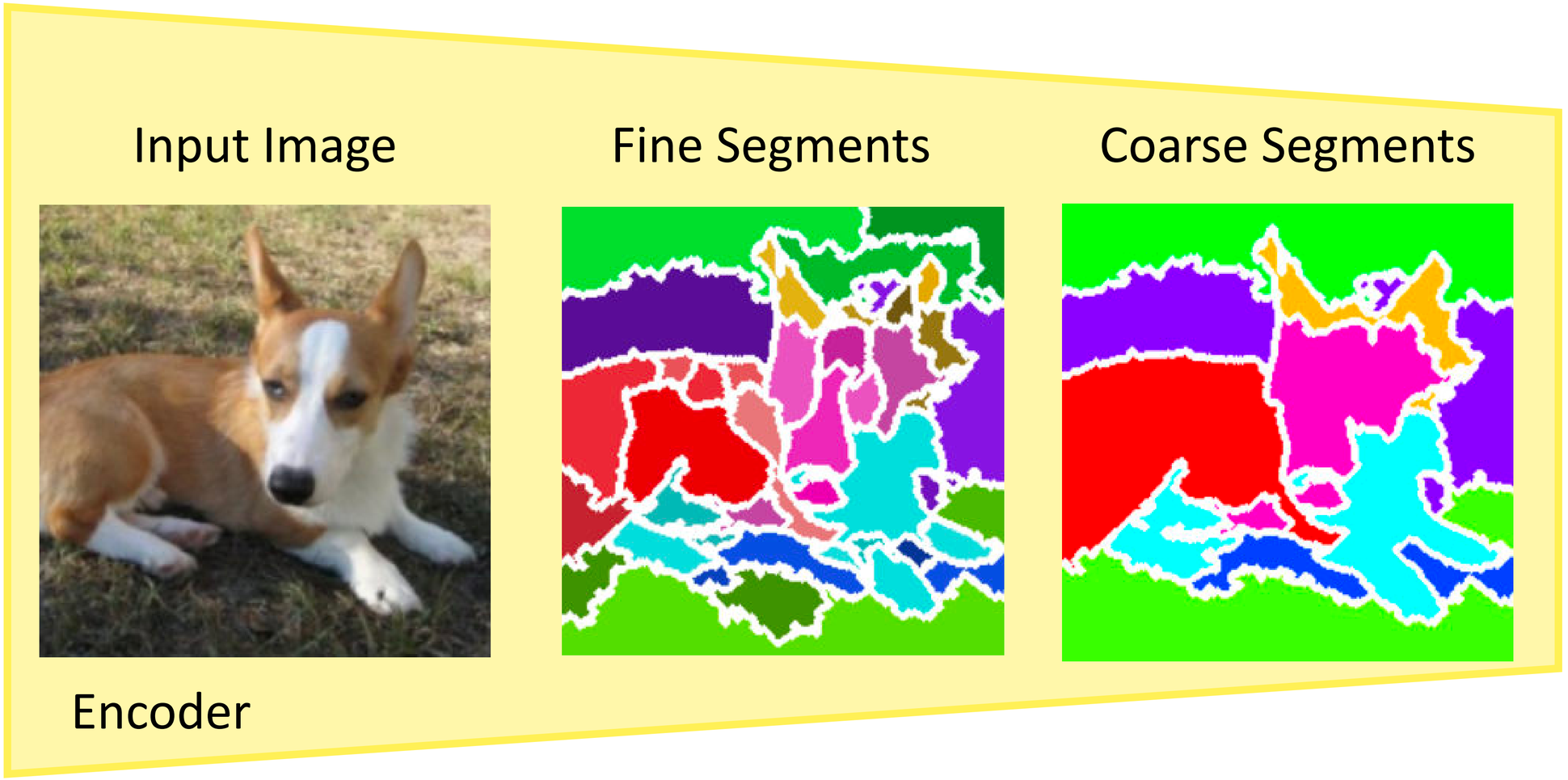


Encoder-only

“Dog”

Ke, Tsung-Wei, Sangwoo Mo, and Stella X. Yu. "Learning hierarchical image segmentation for recognition and by recognition." ICLR (2024)

Overview 1. Encoder for grouping segments with CAST

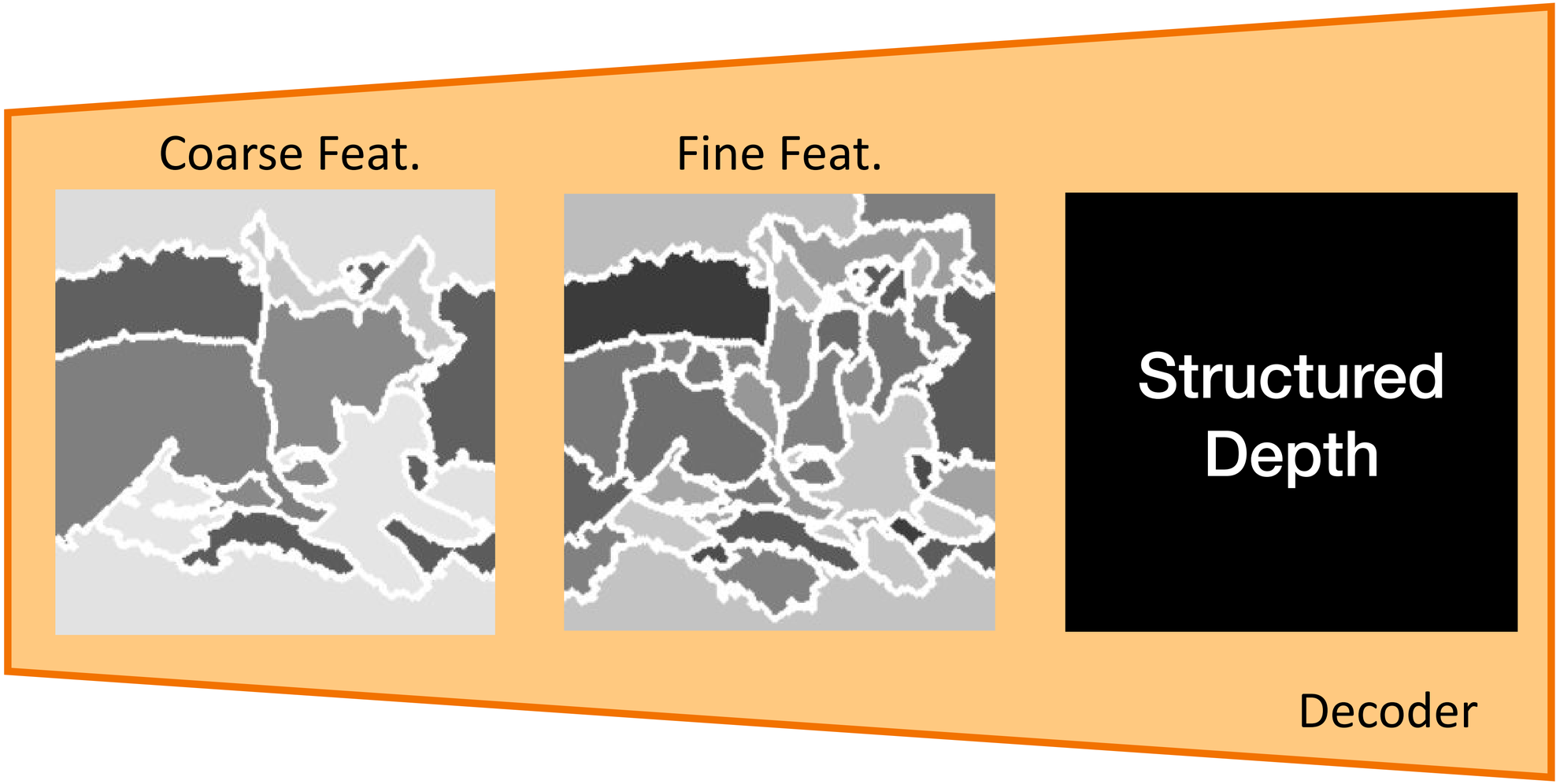


Ke, Tsung-Wei, Sangwoo Mo, and Stella X. Yu. "Learning hierarchical image segmentation for recognition and by recognition." ICLR (2024)

Overview

1. Encoder for grouping segments with CAST

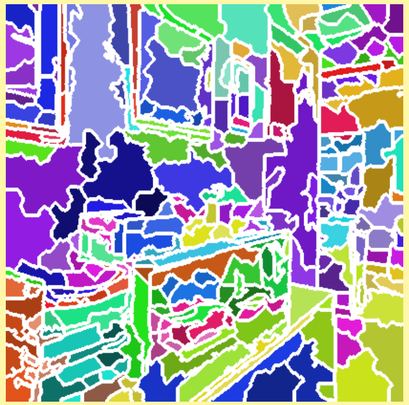
2. Decoding structure through reversing segment hierarchy



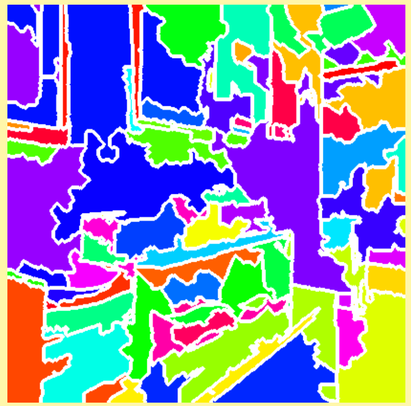
SHED: Modeling Structure by Looping Segmentation



Input



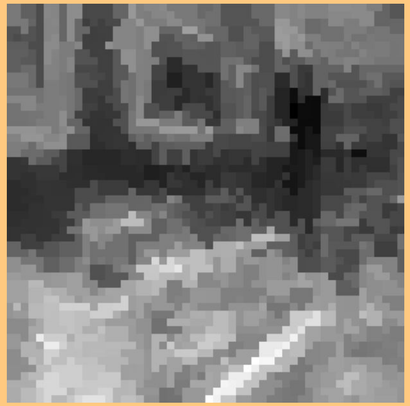
Fine Layout



Coarse Layout



Coarse Feat.



Fine Feat.



Depth Pred.

Encoder

Decoder

Encoder 1. Tokenization

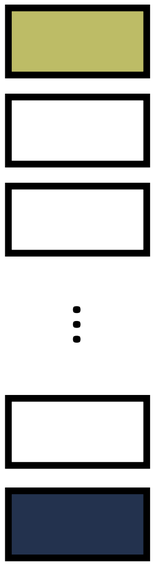
Input Image



Superpixels

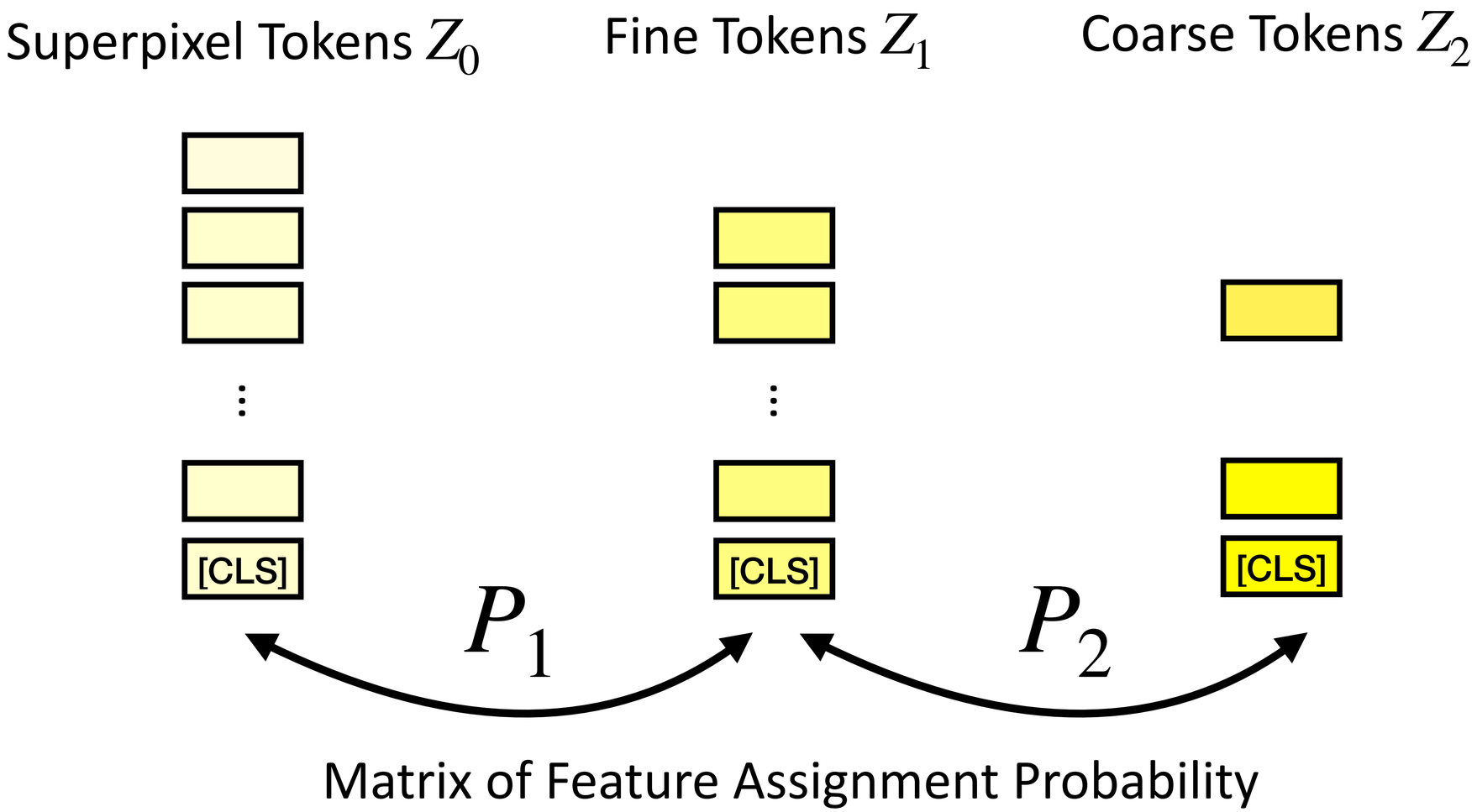


Superpixel Tokens Z_0



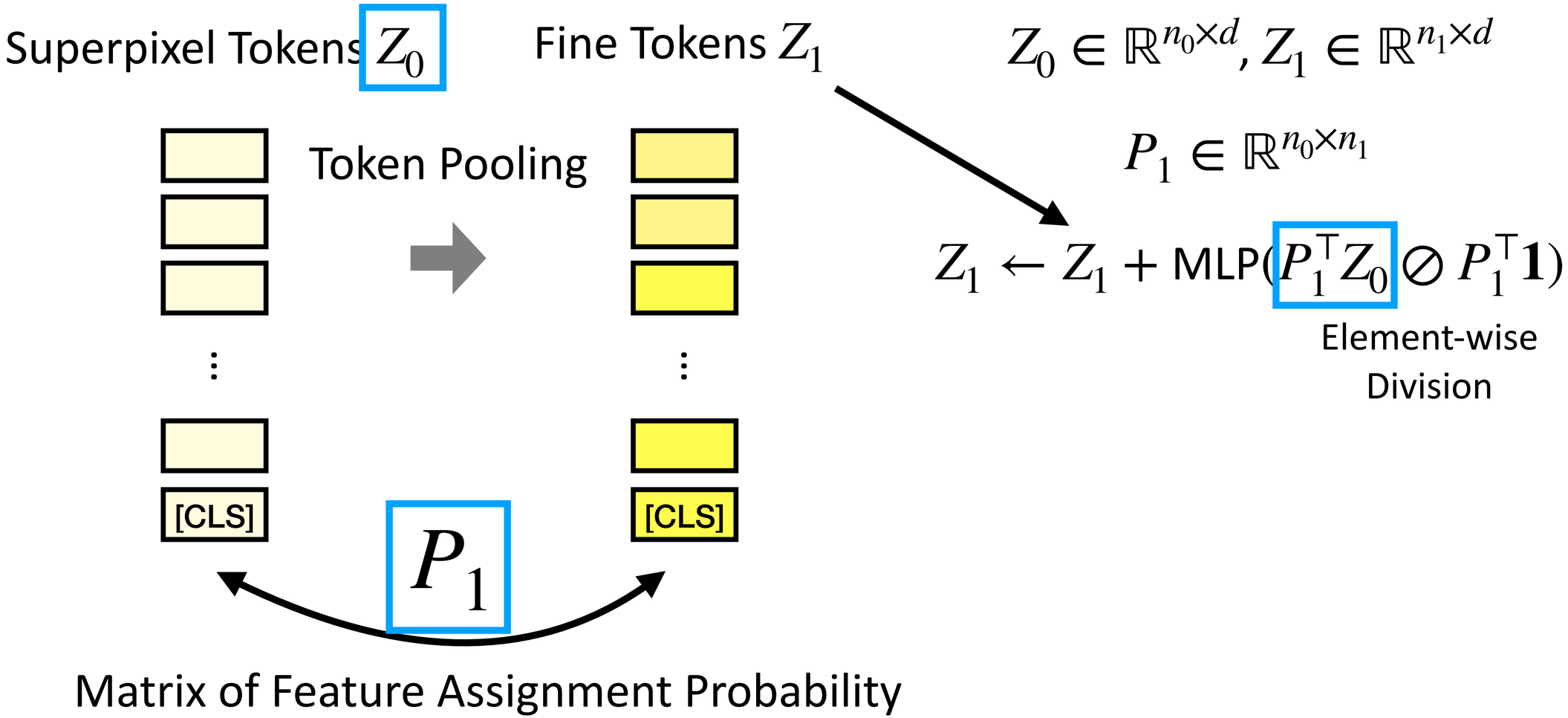
Encoder

1. Tokenization
2. Token hierarchical clustering



Encoder

1. Tokenization
2. Token hierarchical clustering

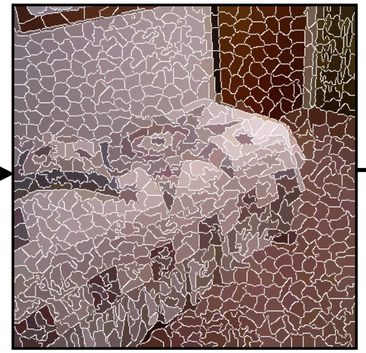


Decoder Decoding structure through reversing segment hierarchy

Input Image



Superpixels



Q. How de we decode this token hierarchy?

Superpixel Token

$Z_0 \in \mathbb{R}^{n_0 \times d}$

Fine-level Token

$Z_1 \in \mathbb{R}^{n_1 \times d}$

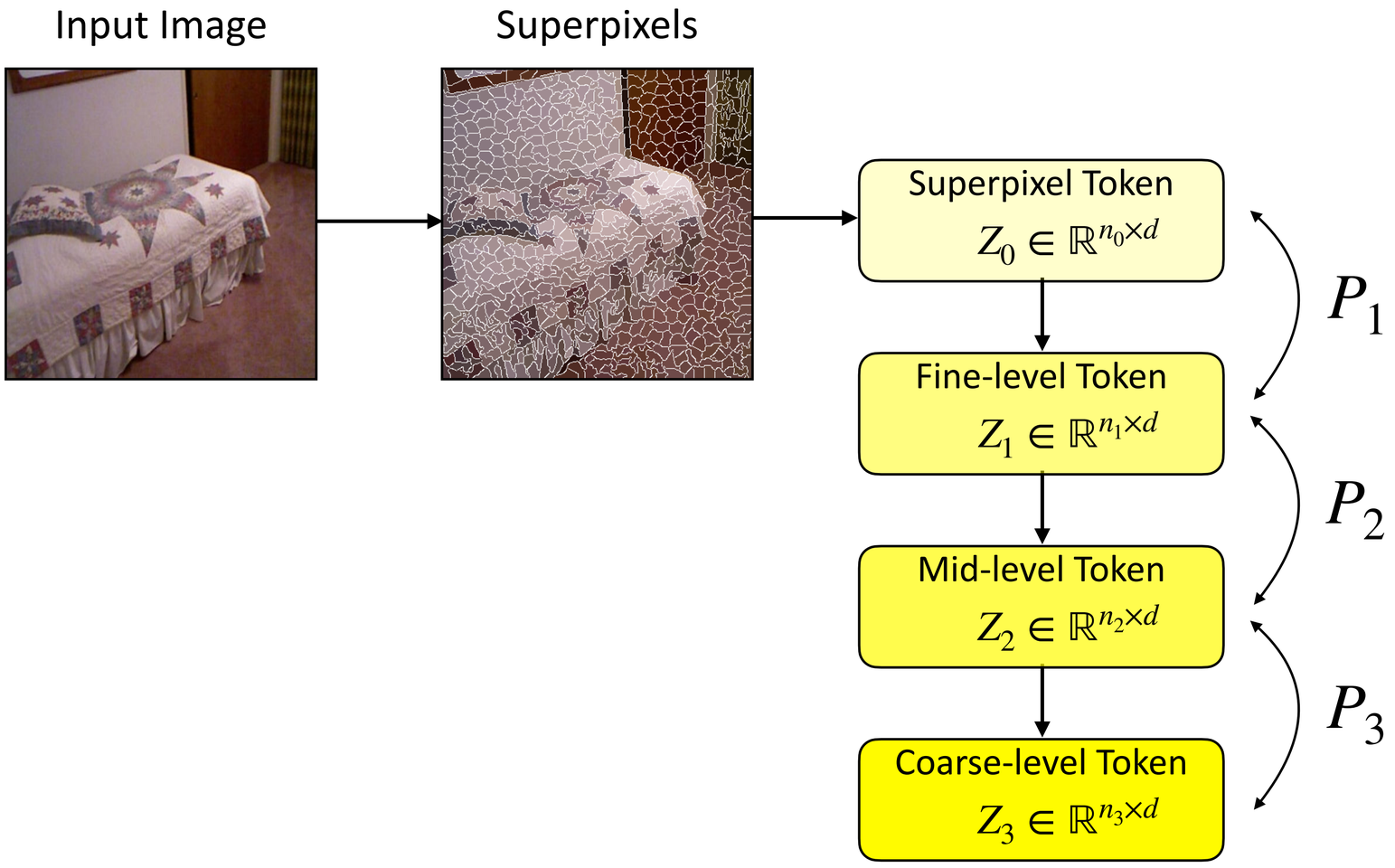
Mid-level Token

$Z_2 \in \mathbb{R}^{n_2 \times d}$

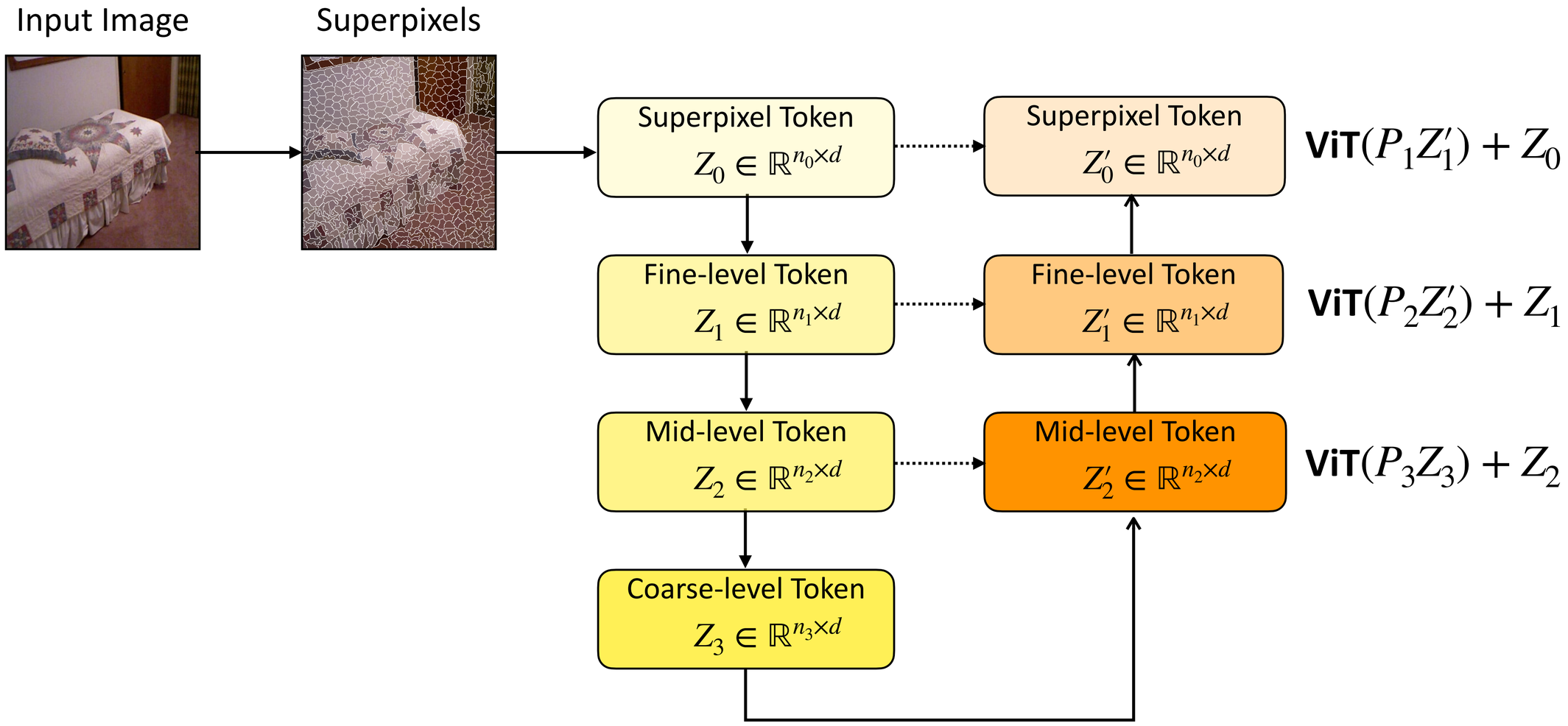
Coarse-level Token

$Z_3 \in \mathbb{R}^{n_3 \times d}$

Decoder 1. Unpooling segment tokens with P

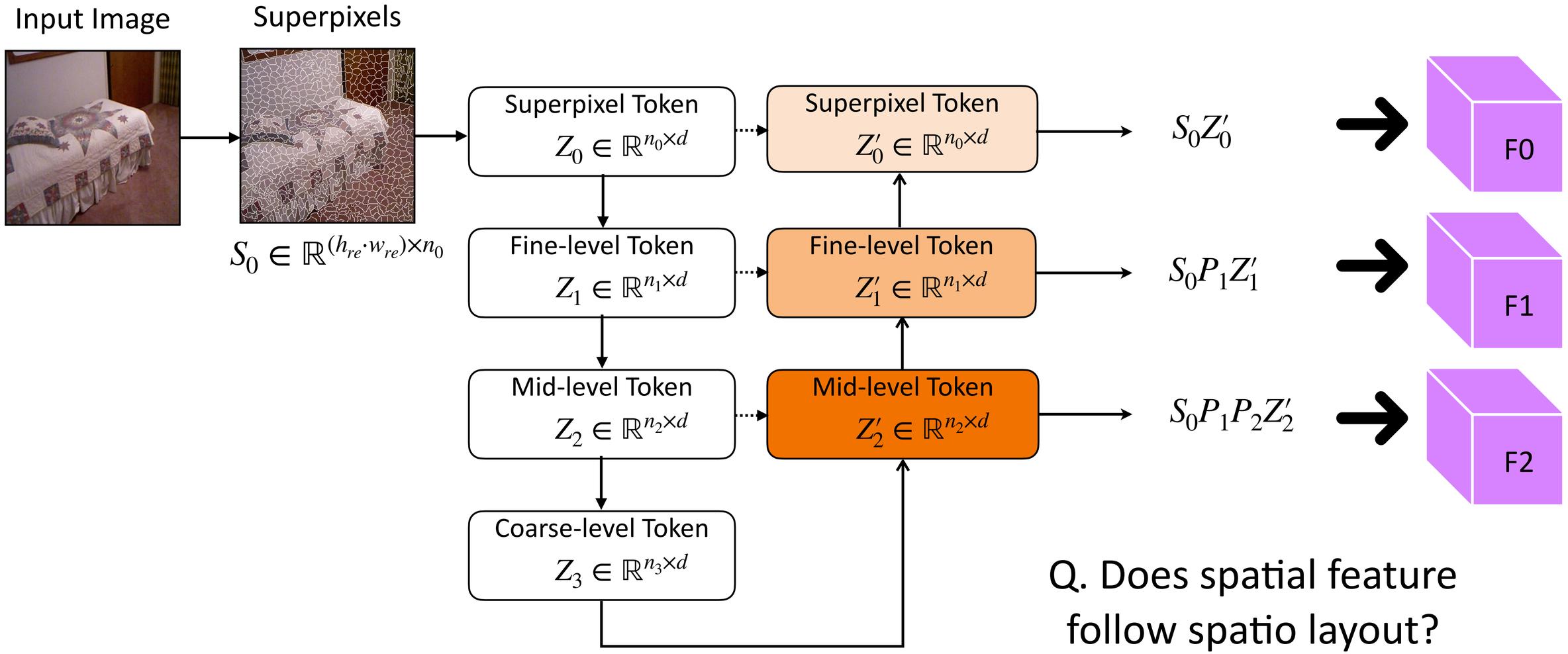


Decoder 1. Unpooling segment tokens with P



Decoder

1. Unpooling segment tokens with P
2. Unpooling spatial features

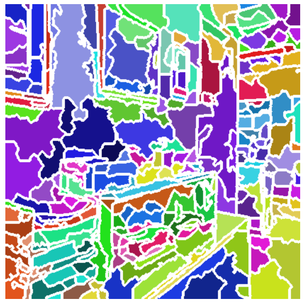


Q. Does spatial feature follow spatio layout?

Decoder

1. Unpooling segment tokens with P
2. Unpooling spatial features

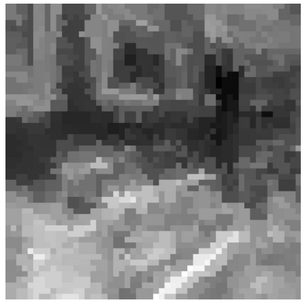
$F_1 \leftarrow S_0 P_1 Z'_1$



$S_0 P_1 \in \mathbb{R}^{(h \cdot w) \times n_1}$

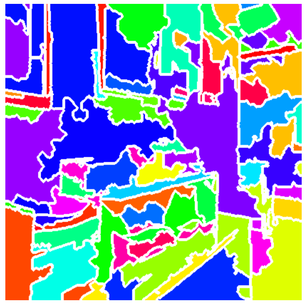
Fine-level Token

$Z'_1 \in \mathbb{R}^{n_1 \times d}$



$F_1 \in \mathbb{R}^{(h \cdot w) \times d}$

$F_2 \leftarrow S_0 P_1 P_2 Z'_2$



$S_0 P_1 P_2 \in \mathbb{R}^{(h \cdot w) \times n_2}$

Mid-level Token

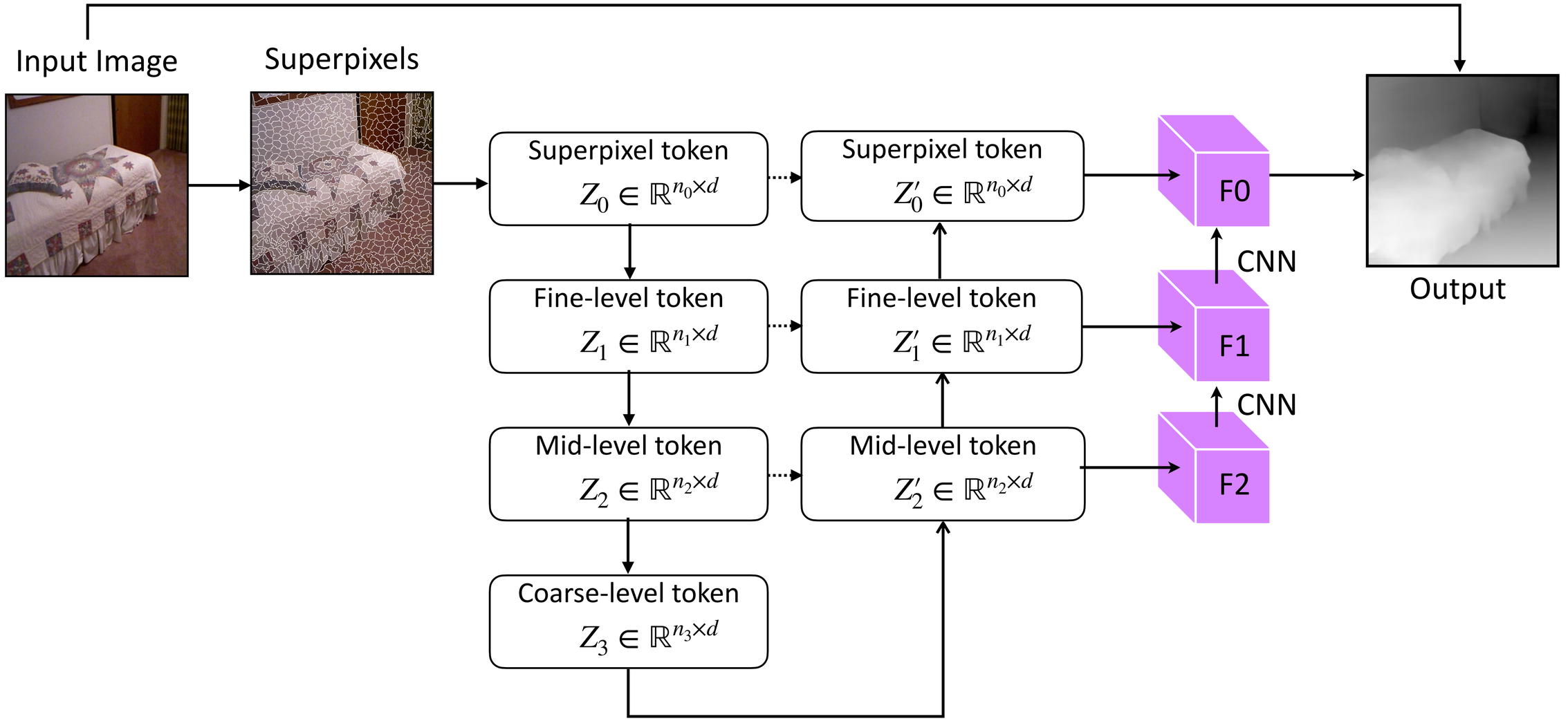
$Z'_2 \in \mathbb{R}^{n_2 \times d}$



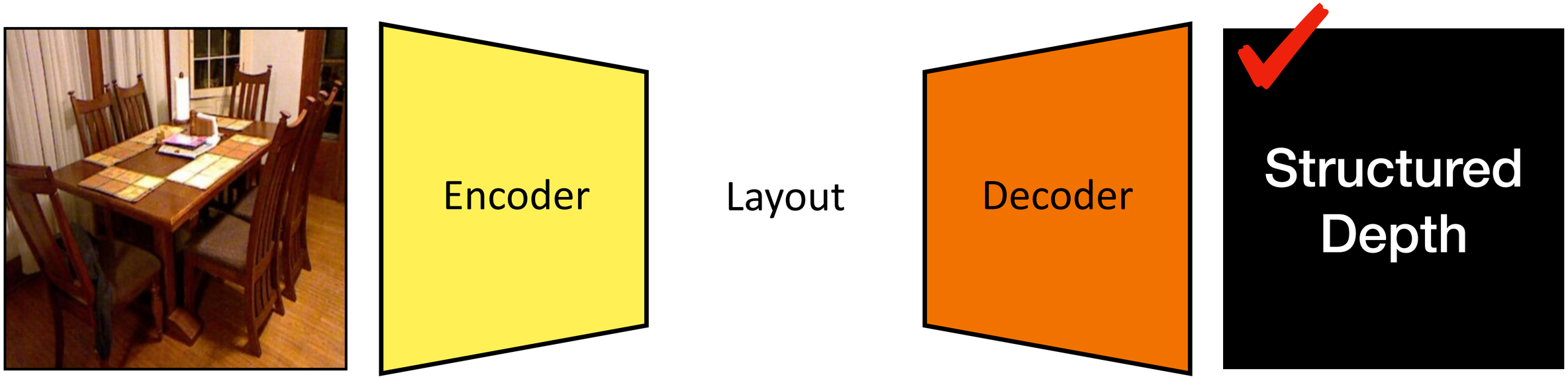
$F_2 \in \mathbb{R}^{(h \cdot w) \times d}$

Decoder

1. Unpooling segment tokens with P
2. Unpooling spatial features

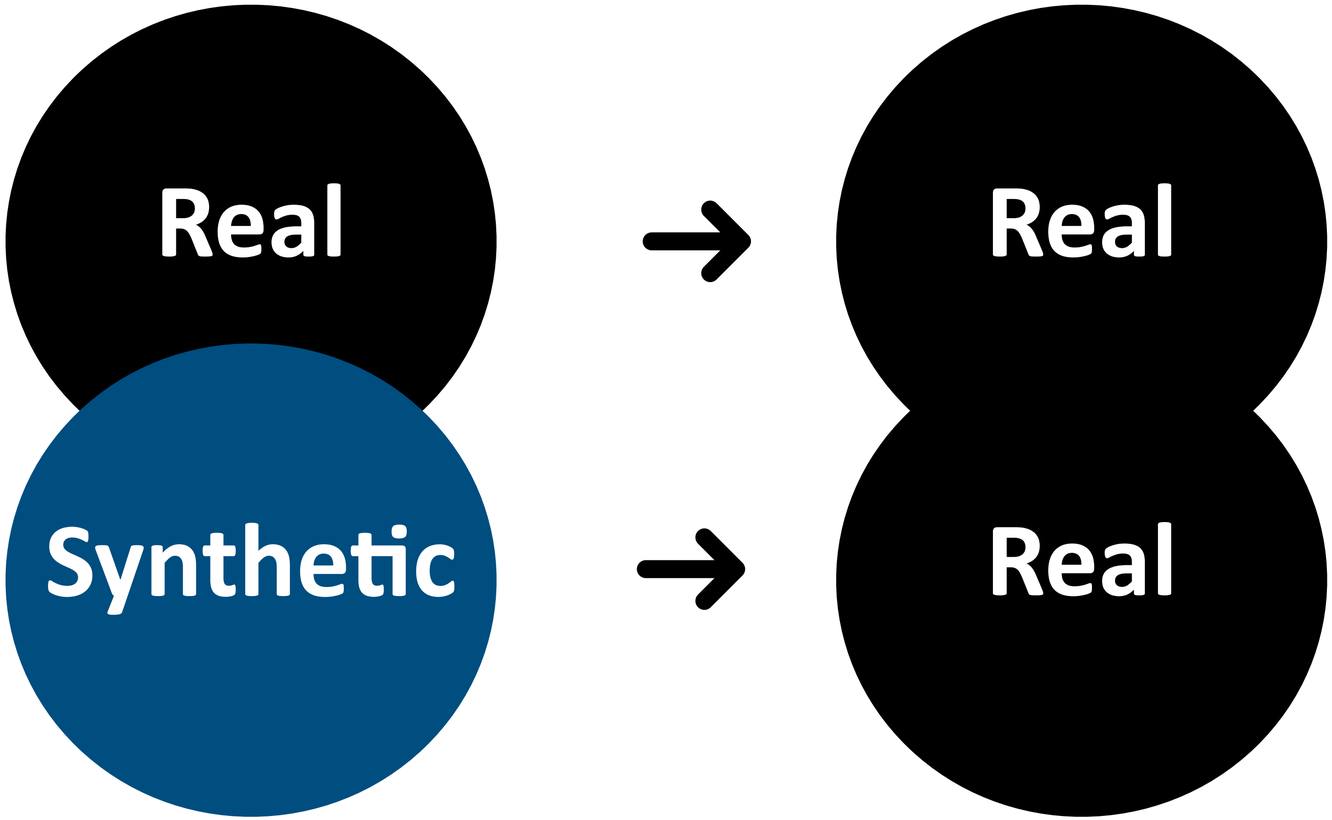


RQ 1. What are the benefits of segment in the loop for MDE?



Clear object boundary / Consistent depth value within object

Experimental Settings

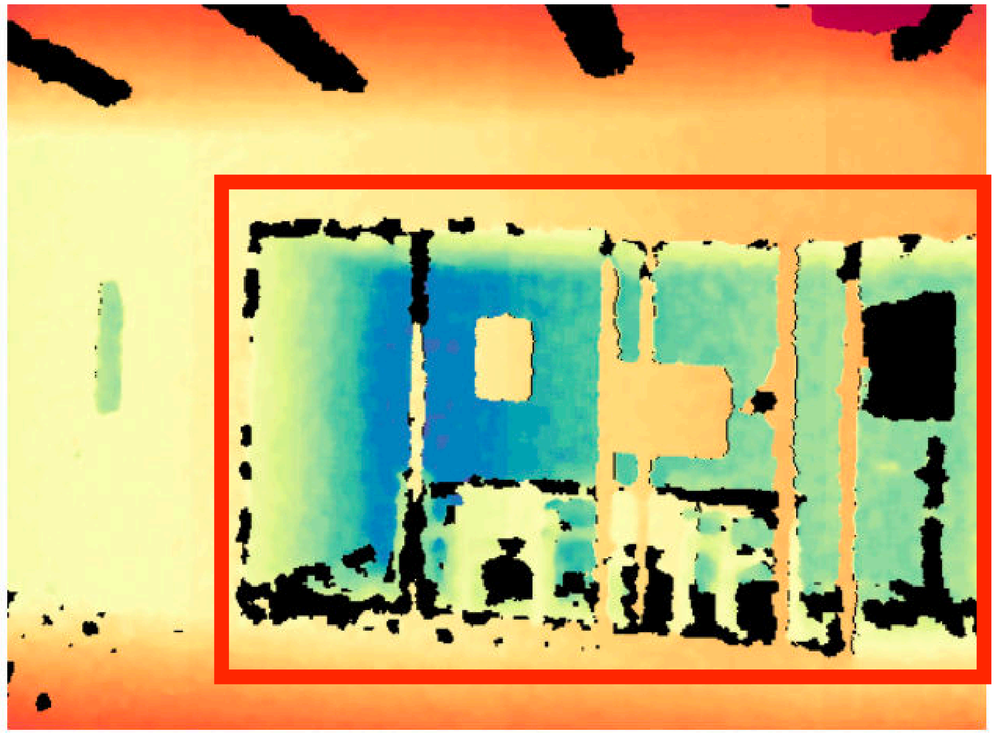


Experiments 1. *Why synthetic data?* - Label noise in GT

Image



GT



Yang, Lihe, et al. "Depth anything v2." NeurIPS (2024)

Experiments

1. *Why synthetic data?* - Label noise in GT

2. *Why synthetic data?* - Coarse depth maps in GT

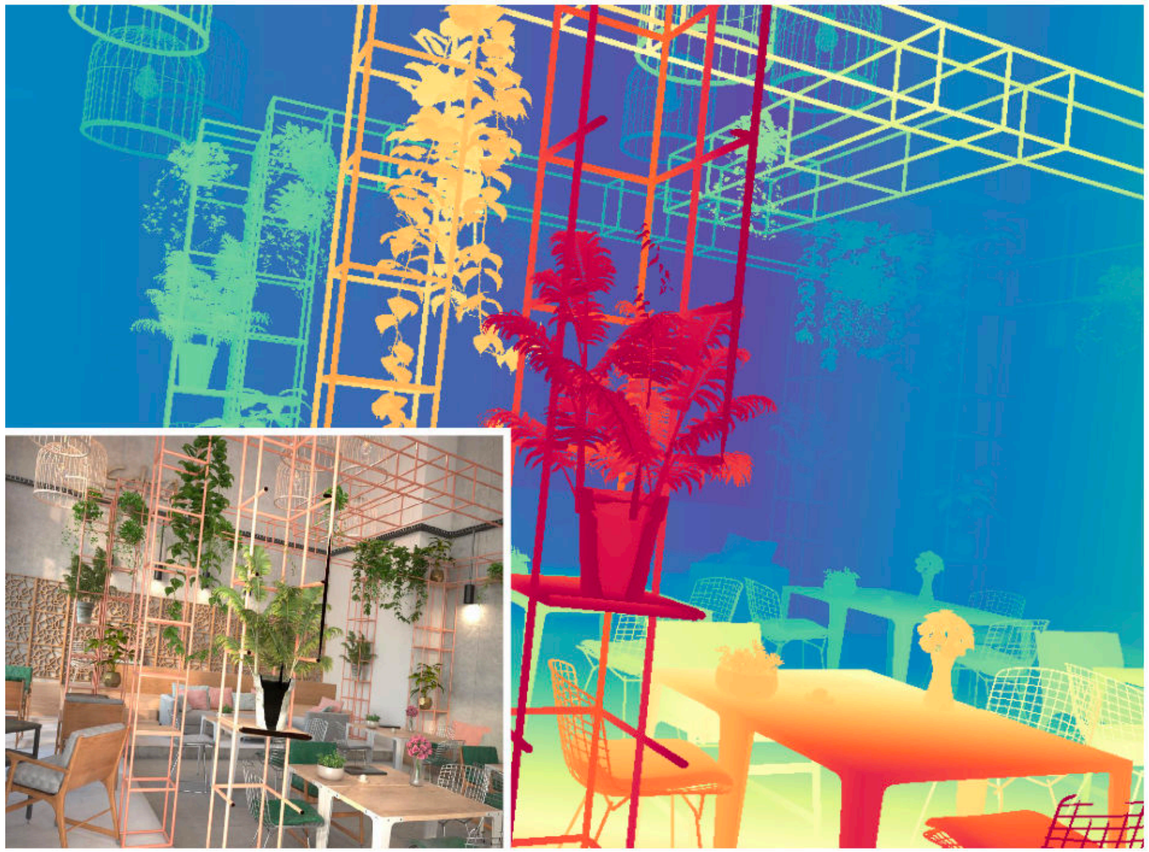




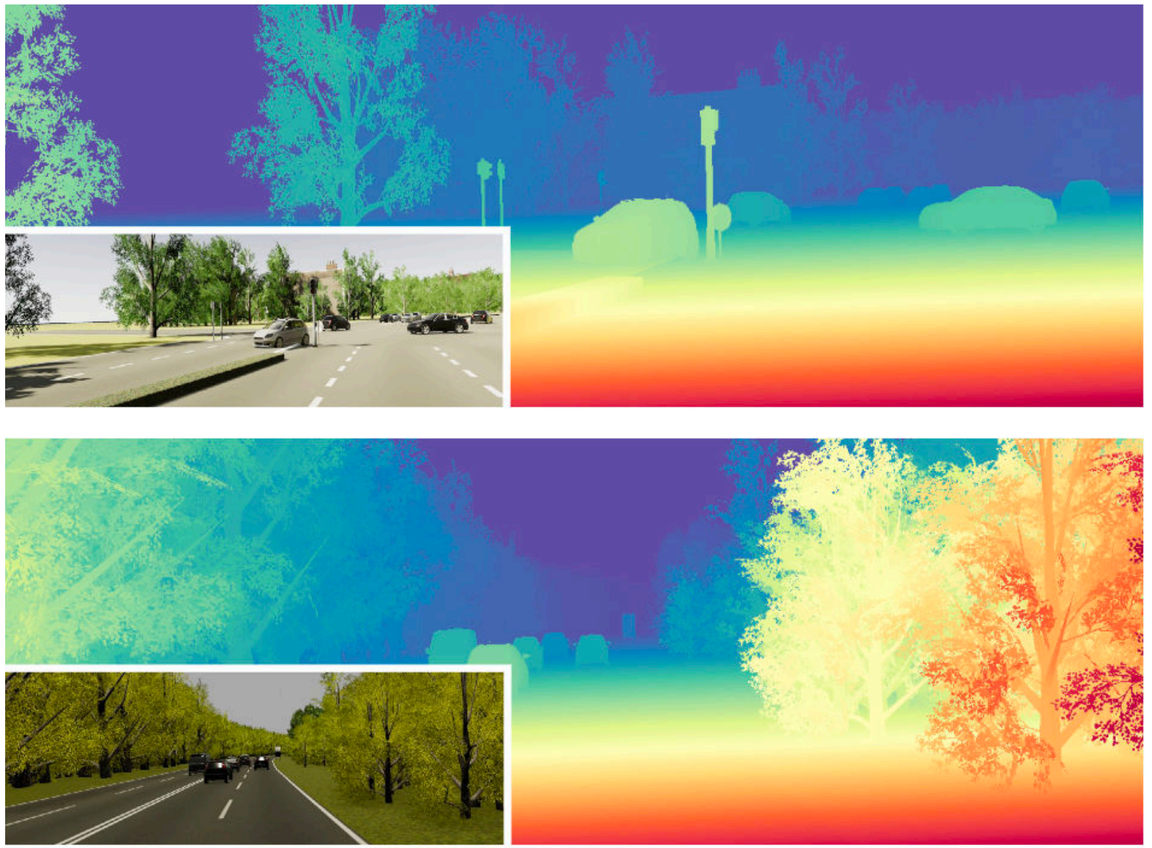
Yang, Lihe, et al. "Depth anything v2." NeurIPS (2024)

Experiments

1. *Why synthetic data?* - Label noise in GT
2. *Why synthetic data?* - Coarse depth maps in GT
3. *High Quality Synthetic data*



HyperSim
(Indoor)



vKITTI
(Outdoor)

Yang, Lihe, et al. "Depth anything v2." NeurIPS (2024)

Sim2Real Performance is Important

Full Supervision with
Synthetic Data



Depth
Foundation
Models



High Quality Depth Map

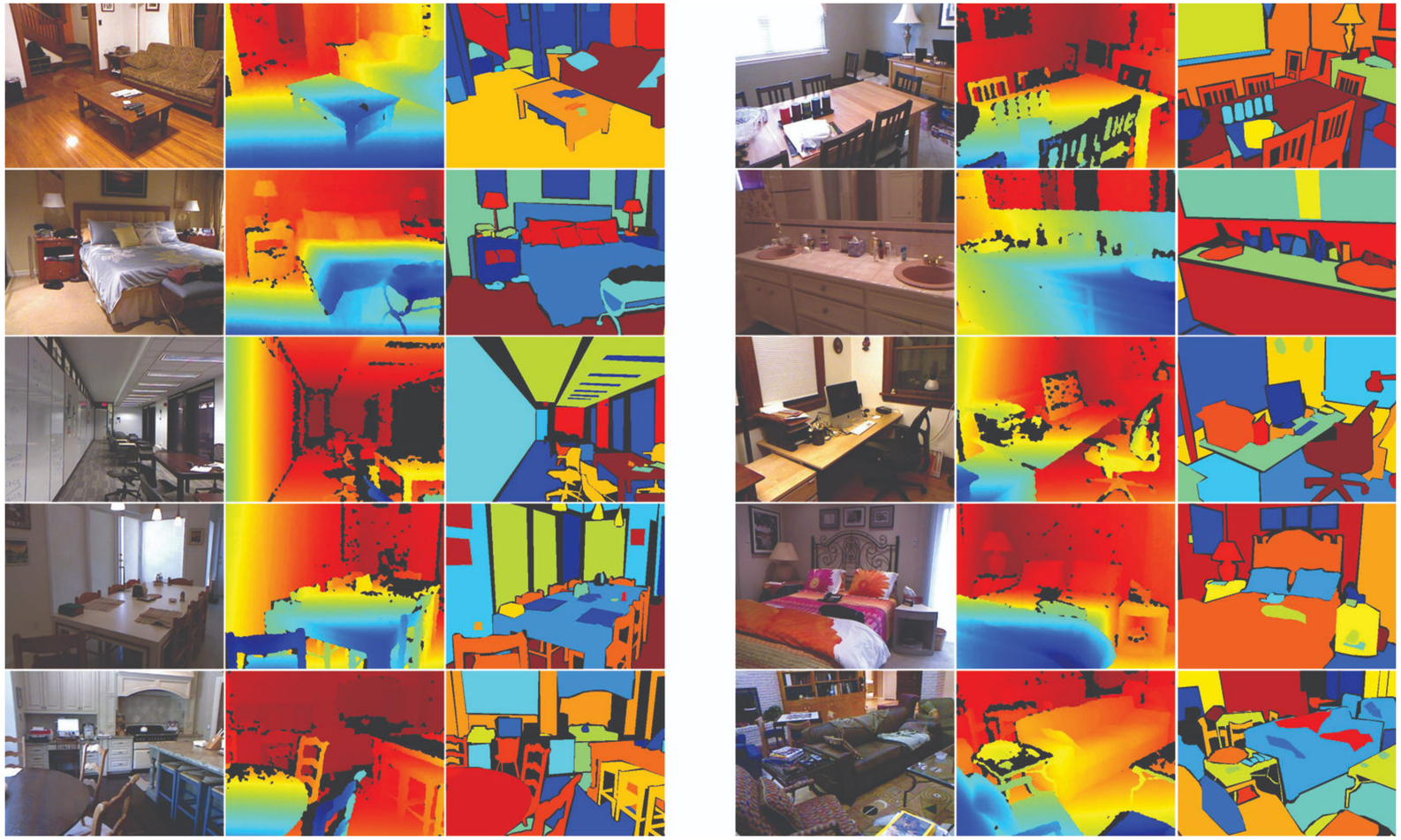
Pre-training with
A Large Scale of Data

Dataset

1. NYUv2 - Real data

Training: 50K RGB and depth pairs with a size of 480 x 640

Test: 654 images

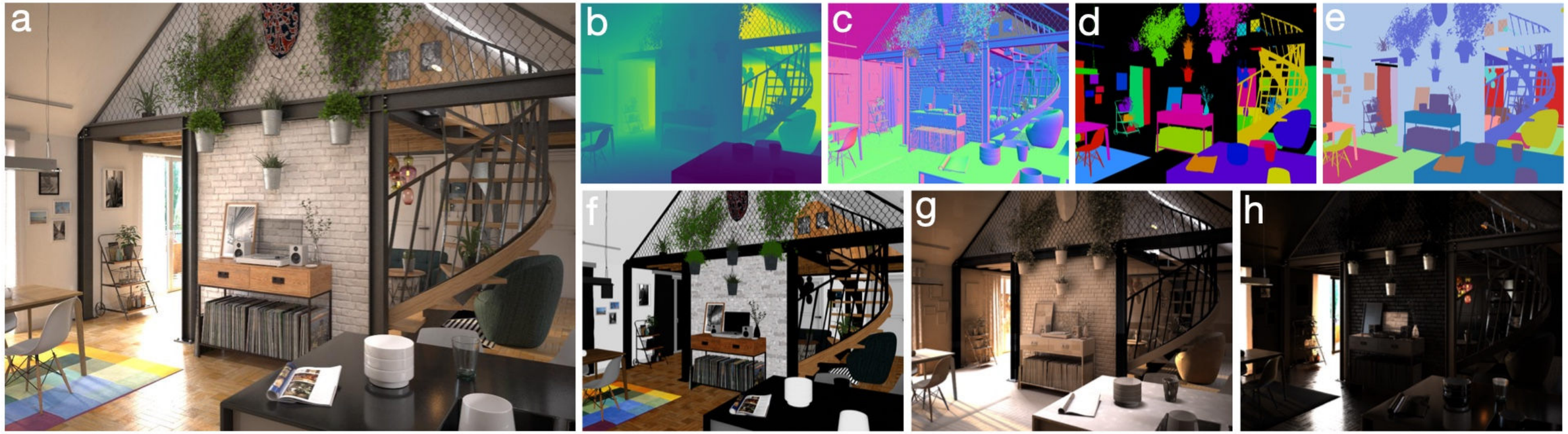


Silberman, Nathan, et al. "Indoor segmentation and support inference from rgbd images" (ECCV 12)

Dataset

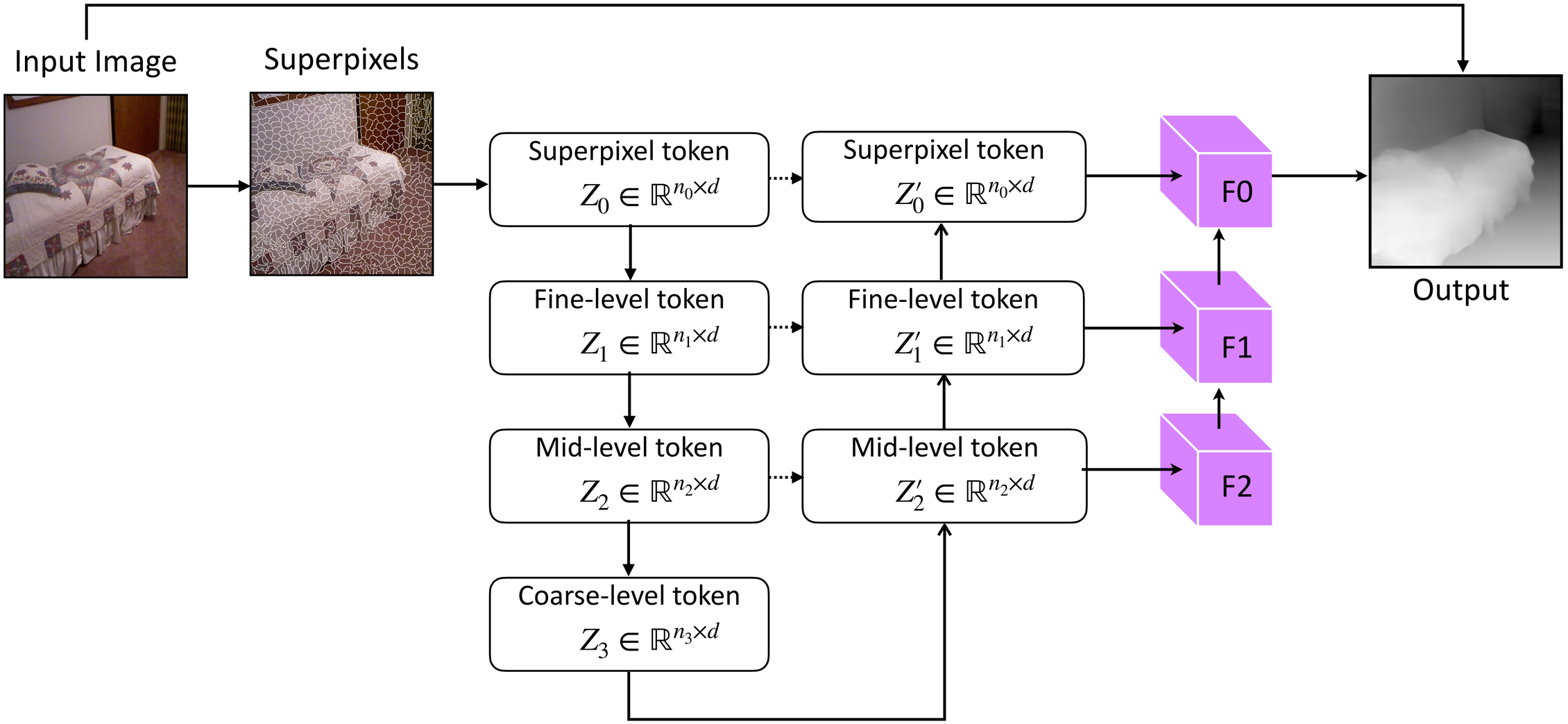
1. NYUv2 - Real world dataset
2. HyperSim - Synthetic data

Training: 59,543 RGB and depth pairs
Validation: 7,386 pairs

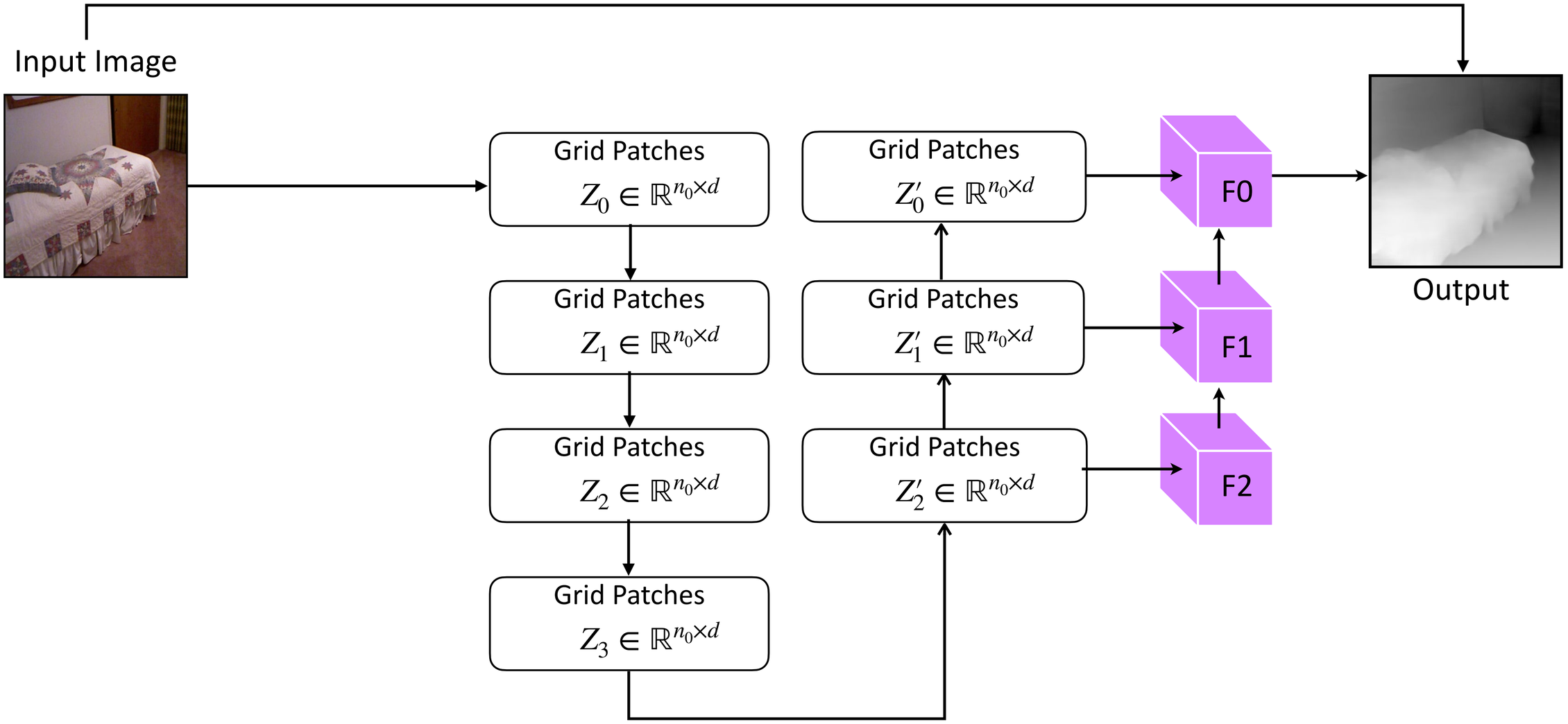


Roberts, Mike, et al. "Hypersim: A photorealistic synthetic dataset for holistic indoor scene understanding." (ICCV 21)

Baseline: DPT



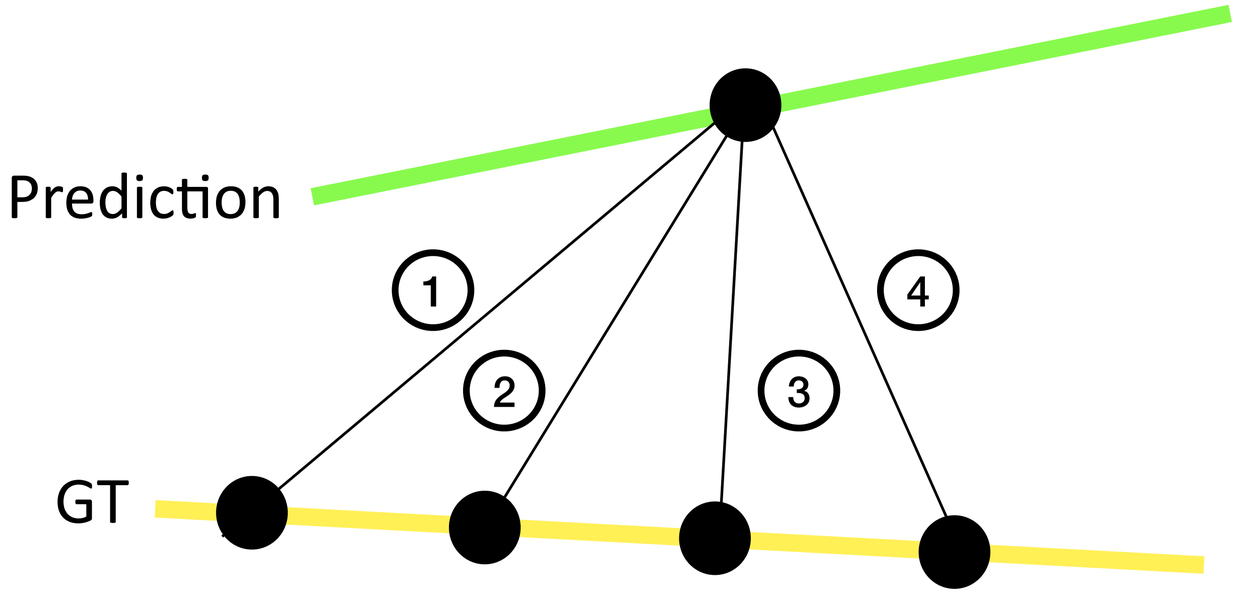
Baseline: DPT



Ranftl, René, Alexey Bochkovskiy, and Vladlen Koltun. "Vision transformers for dense prediction." (ICCV 2021)

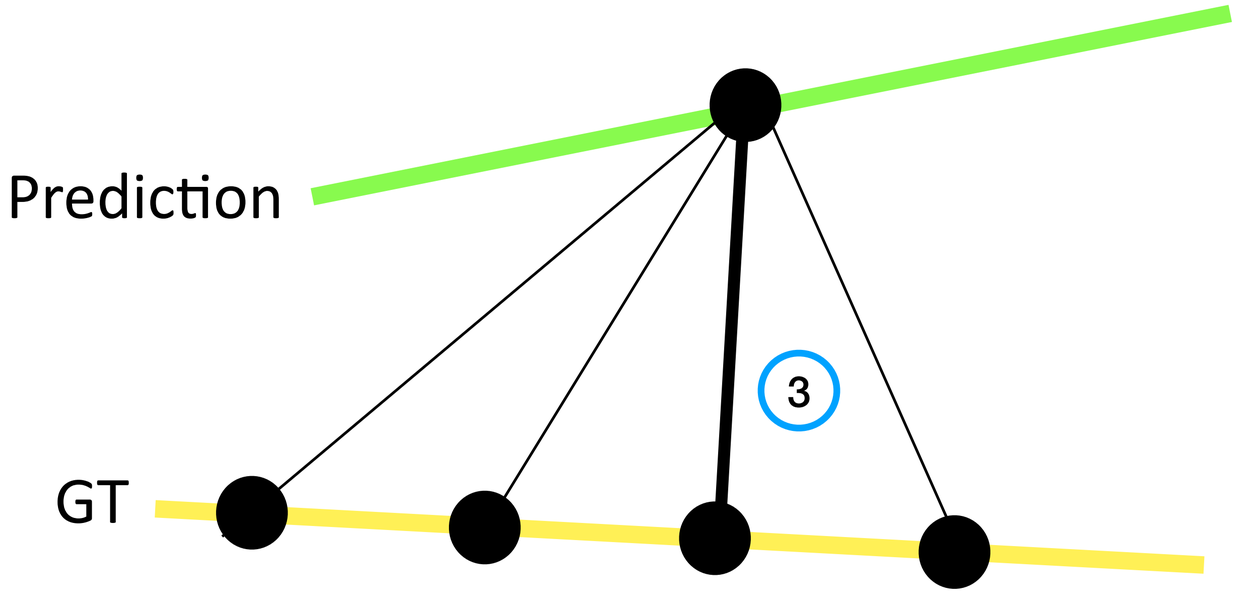
Metric: Occlusion Boundary Error

- **Chamfer distance** between edges extracted from predicted and GT depth map.
- $$\epsilon_a = \frac{1}{|P|}\sum_{p\in P}\min_{g\in G}||p-g||\ ,\ \epsilon_c = \frac{1}{|G|}\sum_{g\in G}\min_{p\in P}||g-p||$$



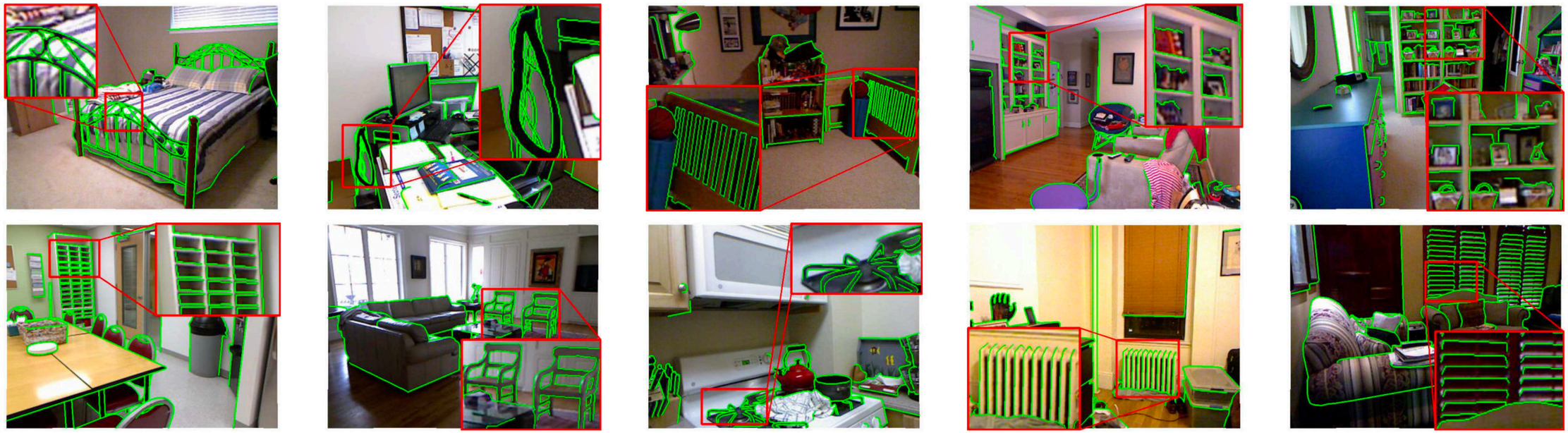
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Occlusion Boundary Benchmark

654 test images manually annotated by human



Samples from NYUv2-OC ++ Dataset

Ramamonjisoa, Michael, et al. "Predicting sharp and accurate occlusion boundaries in monocular depth estimation using displacement fields." (CVPR 20)

Input DPT SHED Ground Truth

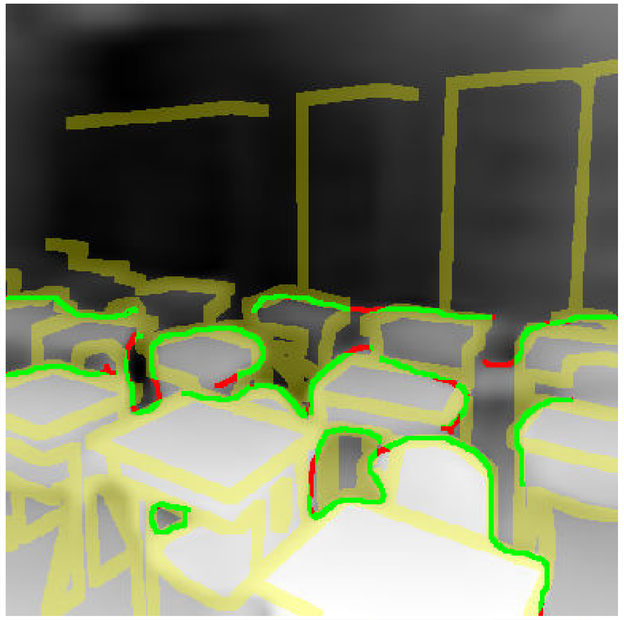


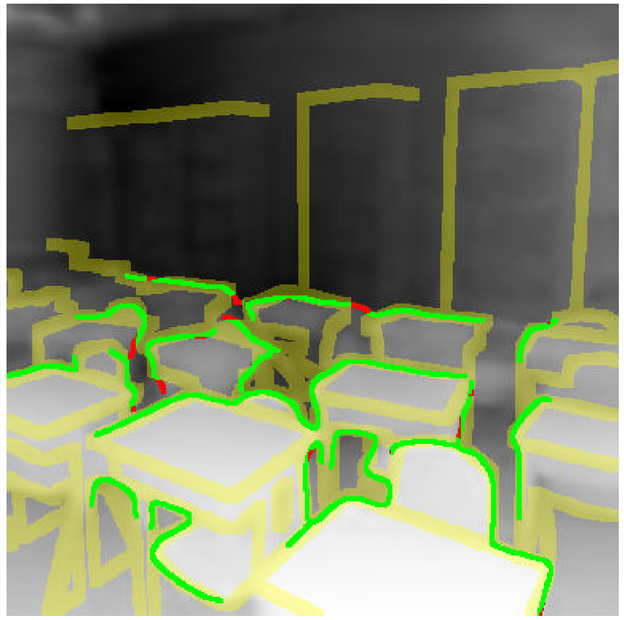
Input

DPT

SHED



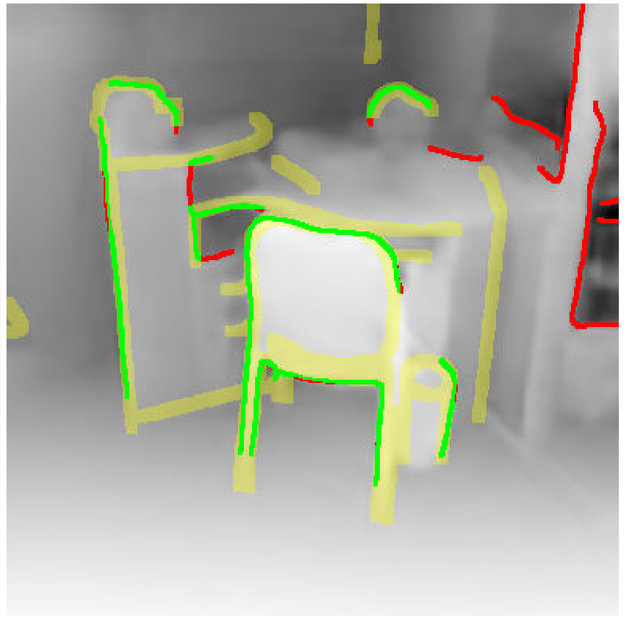




Yellow: GT
Green: True Positive
Red: False Positive







Yellow: GT
Green: True Positive
Red: False Positive

Results

1. SHED improves boundary accuracy

- In-domain evaluation

Method	Prior	Boundary Error (ϵ_a, ϵ_c) ↓
DPT	ImageNet-1K (1M)	6.395 / 1.438
SHED (ours)	ImageNet-1K (1M)	**5.656 / 0.383**

- Sim2Real evaluation

Method	Prior	Boundary Error (ϵ_a, ϵ_c) ↓
Depth Anything 2	LVD-142M (142M)	5.763 / 0.682
SHED (ours)	**ImageNet-1K (1M)**	**5.214 / 0.232**

Metric: Pixel-wise Depth Accuracy and Error

- Depth Accuracy (δ_1) : proportion of depth map where $\max\left(\frac{D_i}{\hat{D}_i}, \frac{\hat{D}_i}{D_i}\right) < 1.25$
- Depth Error (AbsRel) : $\frac{1}{N}\sum_{i=1}^{N}\frac{|D_i - \hat{D}_i|}{D_i}$

Metric: Object-wise Depth Accuracy and Error

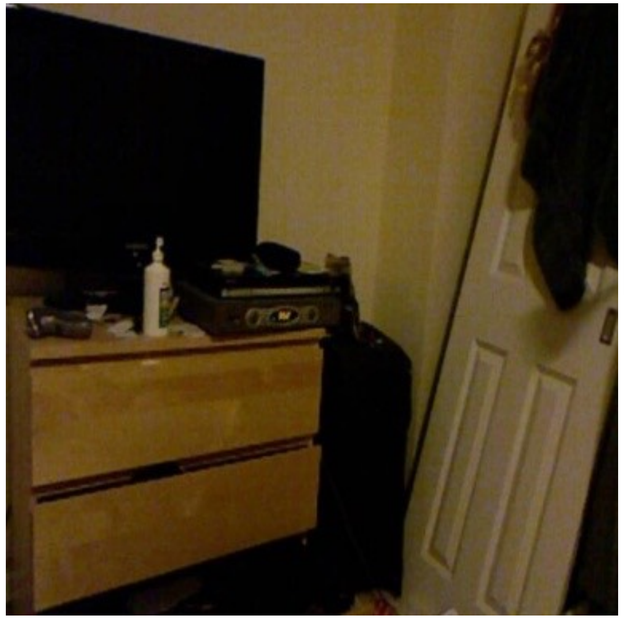
- Object-wise Depth Accuracy (δ_1) : proportion of depth map where $\max\left(\frac{D_i}{\hat{D}_i}, \frac{\hat{D}_i}{D_i}\right) < 1.25$ within segmentation mask
- Object-wise Depth Error (AbsRel): $\frac{1}{N}\sum_{i=1}^{N}\frac{|D_i - \hat{D}_i|}{D_i}$ within segmentation mask

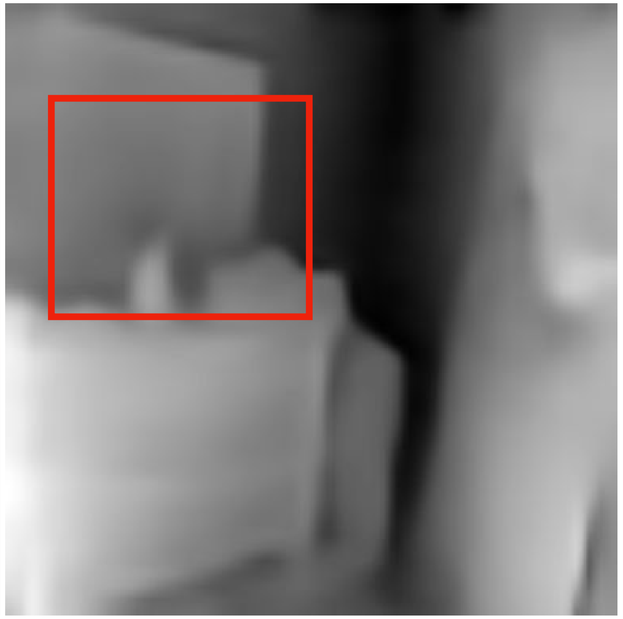
Input Image

DPT

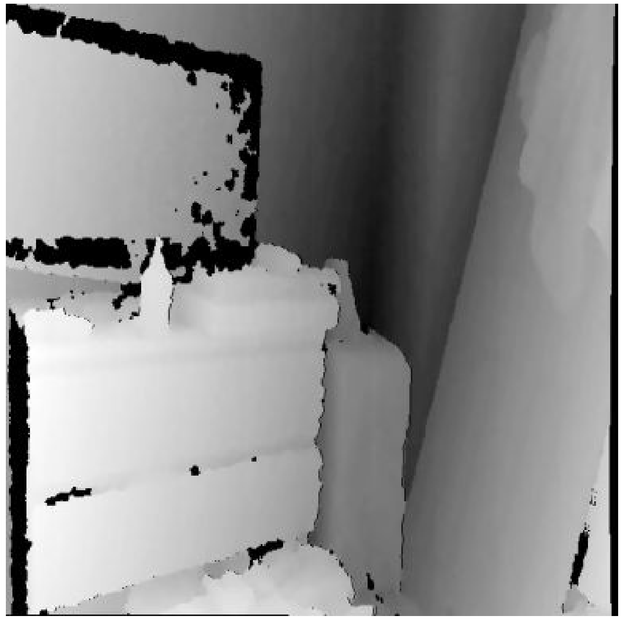
SHED

Ground Truth

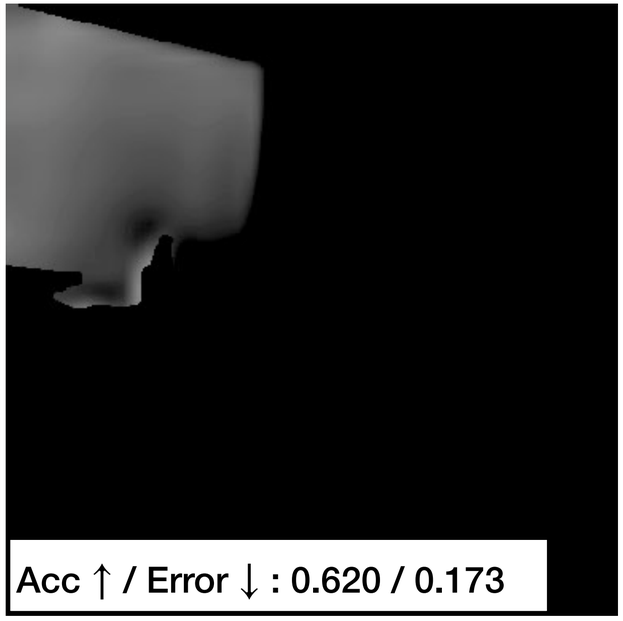


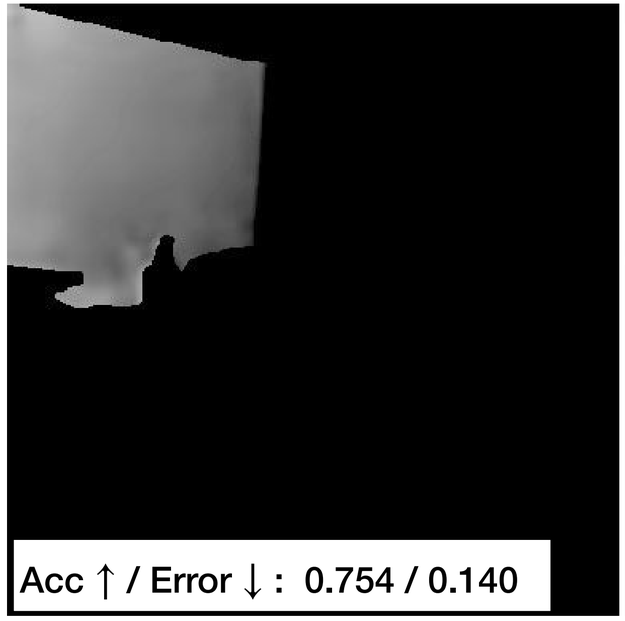














Results

1. SHED improves boundary accuracy

2. SHED improves intra-segment coherence

- In-domain evaluation

Method	Prior	Object-wise Depth Accuracy ↑	Object-wise Depth Error ↓
DPT	ImageNet-1K	0.805	0.141
SHED (ours)	ImageNet-1K	**0.821**	**0.139**

- Sim2Real evaluation

Method	Prior	Object-wise Depth Accuracy ↑	Object-wise Deph Error ↓
Depth Anything 2	LVD-142M	0.573	0.251
SHED (ours)	ImageNet-1K	**0.684**	**0.189**

Results

1. SHED improves boundary accuracy
2. SHED improves intra-segment coherence
3. **SHED achieves strong sim-to-real generalizaton**

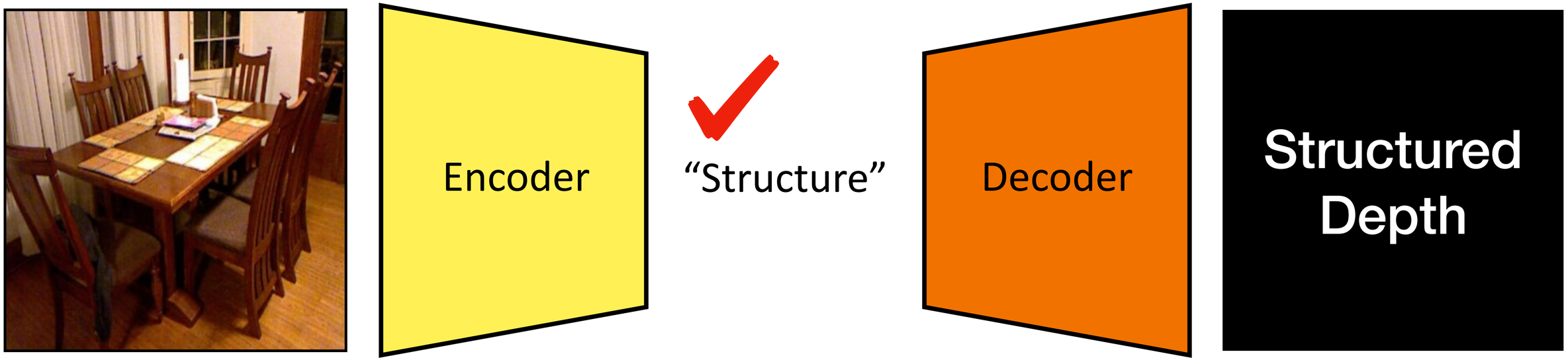
- In-domain evaluation

Method	Prior	Boundary Error ↓	Object-wise Depth Acc ↑	Object-wise Depth Error ↓	Depth Acc ↑	Depth Error ↓
DPT	ImageNet-1K	6.395 / 1.438	0.805	0.141	0.829	0.132
SHED (ours)	ImageNet-1K	**5.656 / 0.383**	**0.821**	**0.139**	**0.842**	**0.131**

- Sim2Real evaluation

Method	Prior	Boundary Error ↓	Object-wise Depth Acc ↑	Object-wise Depth Error ↓	Depth Acc ↑	Depth Error ↓
Depth Anything 2	LVD-142M	5.763 / 0.682	0.573	0.251	0.613	0.729
SHED (ours)	ImageNet-1K	**5.214 / 0.232**	**0.684**	**0.189**	**0.616**	**0.619**

RQ 2. Do internal features represent spatio-layout oriented structure?



Structure-aware Image Retrieval / Depth-aware Segment Hierarchy

Results

1. SHED improves boundary accuracy
2. SHED improves intra-segment coherence
3. SHED achieves strong sim-to-real generalizaton
4. SHED learns layout-aware representation

Query

DPT Retrieval Results



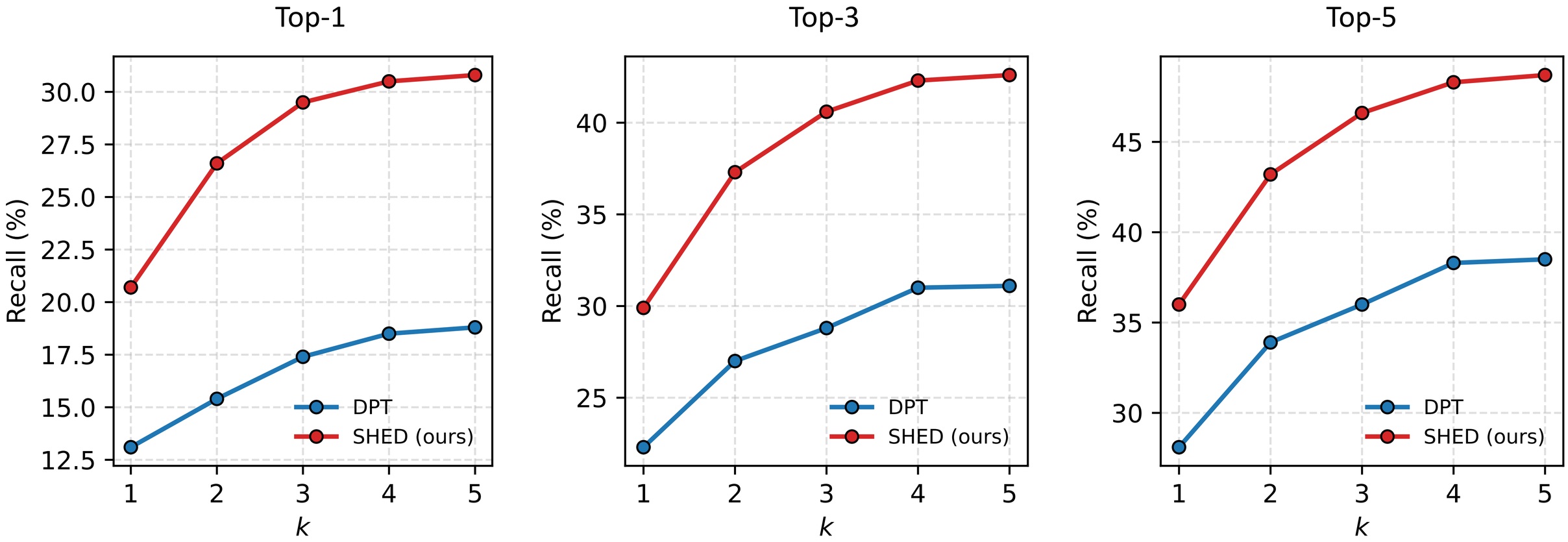
Query

SHED (ours) Retrieval Results



Frame-level Image Retrieval

Top-1,3,5 recall within frame time step k



Results

...

3. SHED achieves strong sim-to-real generalizaton

4. SHED learns layout-aware representation

5. SHED learns depth-aware segment hierarchies

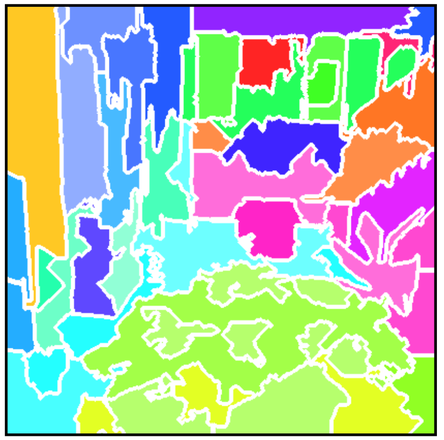
Input Image

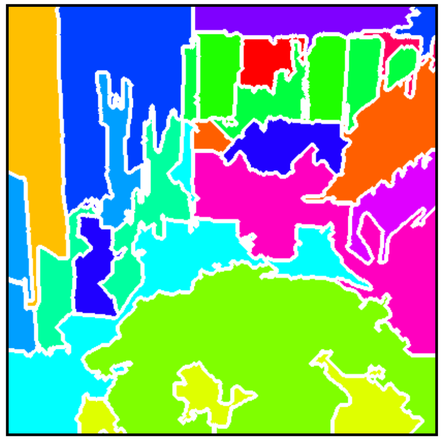
CAST: 64, 32, 16 segments

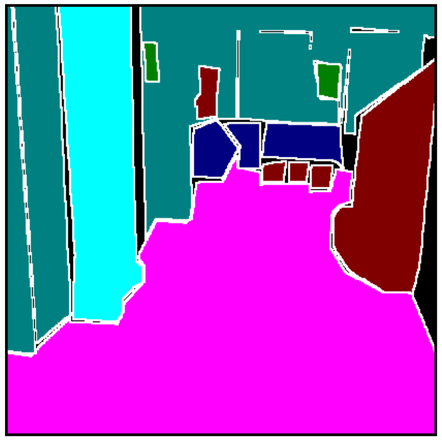
GT Segmentation











SHED: 64, 32, 16 segments

GT Depth









Future Work

- Scaling up SHED



Future Work

- Other dense prediction tasks



https://medium.com/@d.d.tananaev/motion-modeling-in-videos-part-1-optical-flow-68901cf01435

Conclusion

- This research proposes SHED, a novel approach that integrates internal segmentation with depth regression to represent image structure.
- SHED demonstrates superior performance in occlusion boundaries and intra-segment coherence compared to baselines like DPT and foundation models like Depth Anything 2.
- SHED's layout-aware representation significantly outperforms DPT, indicating that its internal segmentation captures the image's structure more effectively.
- The internal segmentation hierarchies learned by SHED are depth-aware, aligning better with the 3D structural components of the scene compared to segmentation methods that uses semantic information.